

Popular Entertainment as Propaganda: Insights from an Online Field Experiment in China

Linan Yao, Vanderbilt University

March 14, 2024

Abstract

Can popular entertainment serve as persuasive propaganda for authoritarian regimes? To address this question, I conducted an online field experiment focusing on China's highly popular propaganda movie *Wolf Warrior*. Participants were provided with one-month paid subscriptions to a major streaming platform in China and randomly assigned to start with different movies, including the propaganda movie. Their political opinions were surveyed post-viewing, and their usage of the platform was tracked over the month to observe viewing behavior. The experiment's findings reveal that *Wolf Warrior* was highly appealing to the audience. Results suggest that it can effectively enhance nationalism, economic perception, system pride, and perceived responsiveness among the majority of viewers. However, its effects vary based on the audience's initial attitudes and can occasionally backfire. Notably, behavioral data suggests a high diversity in entertainment tastes among Chinese audiences that do not respond to the treatment, potentially serving as a buffer against the heavy promotion of authoritarian propaganda through entertainment. My research highlights the potent yet nuanced role of entertainment in authoritarian propaganda, illustrating its significant impact on shaping public opinion and political beliefs, as well as its limitations.

“The easiest way to inject a propaganda idea into most people’s minds is to let it go through the medium of an entertainment picture when they do not realize that they are being propagandized.”

– Elmer Davis, Director of U.S. Office of War Information¹

1 Introduction

From the era of silent films to the age of streaming services, entertainment has always had a secret second life as a messenger of political ideas. In the 20th century, totalitarian regimes such as the Nazis and the Communist regime utilized movies as propaganda tools, a practice also adopted by democracies like the UK and the US during wartime. Key propagandists from these diverse regimes, including Joseph Goebbels from Nazi Germany and Elmer Davis from the US, echoed a similar sentiment regarding the effectiveness of movies in shaping public opinion. (Welch, 2008; Coyne and Hall, 2021). As we have moved into the 21st century, although the iron grip of totalitarian states has given way to “informational autocrats” who employ more subtle means of information manipulation (Guriev and Treisman, 2020), the tactic of integrating propaganda within entertainment has continued in many authoritarian countries, such as China, Russia, Egypt, and Iran (Tolz and Teper, 2018; Knobel, 2020; Barshad, 2022; Soffar, 2022; El Banhawey, 2019; Sinaee, 2023; Ershad, 2016).

As for academic research, scholars have amassed a substantial body of literature on authoritarian propaganda; however, the evidence regarding its effectiveness remains mixed. On one hand, observational studies recognize the media’s significant impact on public opinion (Adena et al., 2015; Voigtländer and Voth, 2015). Yet, these studies often focus on media access as a whole, making it difficult to determine the direct effects of specific propaganda pieces, like propaganda movies. On the other hand, research that focuses on individual propaganda pieces, often through controlled experiments, frequently faces challenges in demonstrating their transformative effects on key political attitudes, such as support for the regime. For instance, Mattingly and Yao (2022) found that nationalistic propaganda could increase anti-foreign sentiment without reducing people’s willingness

¹Koppes and Black (1977).

to protest or altering their perception of government performance. Furthermore, [Huang \(2018\)](#) found that hard propaganda could actually worsen people’s opinions of the regime while reducing their willingness to protest, which suggests that such propaganda may be more about intimidation than persuasion. The varied results of these studies, especially the difficulty in identifying propaganda that effectively bolsters regime support, highlight the critical need for more research to understand how authoritarian propaganda can shape political beliefs and perceptions.

However, the current research on authoritarian propaganda has largely overlooked the use of popular entertainment-based propaganda, such as propaganda movies that achieve market success and are voluntarily embraced by audiences.² This oversight is significant, firstly because of the frequent deployment of such entertainment for propaganda and their purported effectiveness, as outlined in the opening quote. Moreover, this gap becomes even more noteworthy in today’s saturated media environment, where captivating and maintaining viewers’ attention is incredibly challenging. The ability of popular entertainment to engage audiences is not just important for real-world influences; it could be crucial for achieving persuasive outcomes. Overlooking the potential of such content as vehicles for authoritarian propaganda risks underestimating the ability of these regimes to shape public opinion.

The empirical motivation for this paper was the resurgence of propaganda movies in China, which presents an opportunity for revisiting the timeless theme of popular entertainment as propaganda. Since transitioning from a totalitarian system to a market economy, the government no longer maintains absolute control over people’s consumption choices in China. As a result, entertainment-based propaganda was marginalized in the market for a long time. However, there has been a recent revival of this genre, particularly in theaters. Notable examples include the *Wolf Warrior* series (2015, 2017). Produced in collaboration with the military, the *Wolf Warrior* series portrays a confident and assertive nation and has gained enormous popularity with domestic audiences. Perhaps the best indication of the movie’s impact is its influence on the informal naming of China’s increasingly assertive foreign policy as the “Wolf Warrior” policy. Another example is *Battle at Lake Changjin* (2021), a Korean War movie set against the backdrop of escalating China-U.S. relations, which broke box office records and sparked a social media trend where people ate frozen potatoes to

²While some forms of entertainment, like overtly laudatory poems or segments from anti-Japanese TV dramas, have been analyzed, these do not represent the more popular entertainment known for its market appeal. ([Mattingly and Yao, 2022](#); [Huang, 2018](#))

empathize with the hardships faced by Chinese soldiers in the Korean War.

Despite the impressive box office success and audience acceptance of these propaganda movies, a fundamental question remains: Do they persuade? Can such widely-consumed, regime-sponsored entertainment effectively win over the hearts and minds of audiences in favor of the regime?

To answer this question, I conducted an online field experiment using *Wolf Warrior*, a 2015 action thriller with a patriotic theme, as the treatment. In my experiment, participants received accounts with a one-month subscription to a popular streaming platform. Meanwhile, they are randomly assigned to one of three groups: the forced-exposure group, the placebo control group, and the free-choice group. The forced-exposure group is directed to watch a propaganda movie, while the placebo control group views a non-political movie. The free-choice group has the liberty to choose between propaganda and non-propaganda movies. After viewing the movies, participants are surveyed on their key political opinions. Their subsequent viewing history on the platform is tracked over the month. 362 participants were enrolled via a Chinese crowdsourcing website.

This experiment is novel in several respects. Firstly, the experiment site is iQIYi, the most popular streaming platform in China. This setting closely mirrors the natural environment in which people typically consume movies online, thereby enhancing the experiment’s “fieldness.” Secondly, by inviting participants to watch an entire movie rather than just a clip, the experiment provides a more potent dosage of information than many previous studies. Lastly, by monitoring people’s real-life consumption choices over a month, I gather behavioral data on consumer behaviors that speaks to the movie’s effects on long-term exposure to propaganda.

The results from my experiment suggest that propaganda movies effectively enhanced nationalism, economic perception, pride in one’s political system (henceforth termed “system pride”), and perceived responsiveness, provided the participants did not harbor an ex-ante aversion to propaganda movies. However, the use of propaganda movies appeared to have backfired among a small portion of the population with a preexisting distaste for propaganda.³ The treatment did not influence the willingness to protest for either group.

This study also reveals that viewing a propaganda movie primarily evoked positive emotions, with pride being the most dominant. Interestingly, the emotional response pattern did not vary between those who have or lack a pre-existing aversion to propaganda movies, although the intensity of these

³The analysis of this heterogeneous treatment effect was pre-registered.

emotions did differ. In the free-choice group, a majority of participants favored propaganda movies over the non-political alternative, underscoring the appeal and competitiveness of contemporary propaganda films. Regarding real-life consumption choices, the study suggests that even though the propaganda film was well-received, it did not establish a sustained preference for similar content. Evidently, tastes concerning entertainment were notably diverse.

This study makes three main contributions to the existing literature. It contributes to the propaganda literature, which centers around news, by highlighting the role of popular entertainment as a powerful persuasive tool in authoritarian propaganda. These popular entertainment products are not only genuinely popular, but also shape the majority of the viewers' opinions in favor of the state. Such a tool also suffers from two key limitations, such as the possibility of backfire among the wrong audience and the inability to foster a long-term taste.

Additionally, this research enhances our understanding of the underlying mechanisms through which propaganda can lead to unintended consequences. Although the propaganda film successfully evoked positive emotional responses across diverse viewers, individuals with varying pre-existing opinions exhibited distinct patterns in updating their beliefs. This suggests that the updating of opinions does not solely follow emotional experiences but is instead mediated by a cognitive process.

Finally, the behavioral data from the study reveals a relatively low level of private consumption of propaganda, compared to its substantial market presence in theaters, which serves as a testament to the state's ability to cultivate a favorable market environment for propaganda. Nonetheless, the remarkable viewership achieved by the *Wolf Warrior* series, including 120 million theater viewings, stands out. Considering the significant influence of such productions on public opinion, as suggested by the findings of this research, it becomes evident that entertainment-centric propaganda can wield considerable power in shaping public sentiment and political culture in China.

The rest of the paper is structured as follows: First, I lay the theoretical groundwork and establish expectations based on past research. Then, I present my pre-registered experimental design, followed by a description of the data collection process and the sample used. Next, I discuss the experimental results. Finally, the paper concludes with a summary and potential avenues for further research.

2 Entertainment as a Tool of Authoritarian Propaganda

The 20th century witnessed the emergence of cinema as a powerful medium, and authoritarian regimes were quick to grasp its potential. Adolf Hitler and his propaganda minister, Joseph Goebbels, held a profound fascination for film, leading to the swift nationalization of the German film industry under Nazi rule (Welch, 2008). Unlike newspapers or radio, which often prompted greater scrutiny from audiences, cinema was associated primarily with leisure and entertainment, making it a particularly effective tool for disseminating propaganda (Welch, 2008, p. 56.). In particular, Goebbels noted the power of insidious propaganda within popular entertainment in his diary, stating that “as soon as propaganda as a tendency, as a characteristic, as an attitude, remains in the background and becomes apparent through human beings, then propaganda becomes effective” (p. 57).

Do entertainment products effectively serve as persuasive tools for authoritarian regimes? While anecdotal evidence may suggest they do, rigorous academic research has yet to establish entertainment as a direct catalyst for significant shifts in political opinions within authoritarian contexts. This lack of definitive proof aligns with the broader debate concerning the overall persuasive effectiveness of authoritarian propaganda, as scholars have found mixed evidence for it. On the one hand, observational research on the overall impact of media platforms suggests that media can influence individuals’ political views and behaviors (Adena et al., 2015; Voigtländer and Voth, 2015; Barber and Miller, 2019; Wang, 2021a; Gagliarducci et al., 2020; Peisakhin and Rozenas, 2018; Kern and Hainmueller, 2009), without pinpoint entertainment products as the primary driver for significant political change. Meanwhile, experimental studies that specifically examine the isolated effects of individual propaganda pieces have found more limited evidence of the transformative power of propaganda, especially concerning core political attitudes like support for war or support for authoritarian regimes (Hovland, Lumsdaine, and Sheffield (1949); Mattingly and Yao (2022)).

Part of the reason for these mixed results might lie in the fact that not every piece of propaganda is created equal, even within entertainment products. Some scholars, such as Wedeen (1998); Huang (2018); Little (2017), have shown that exaggerated and rigid propaganda can serve as a tool of intimidation, whereby the regime signals its strength, deterring people from protesting on the streets while exacerbating their negative attitudes towards the state. In contrast, other scholars

have studied softer forms of propaganda and shown how these products can play a bigger role in persuasion by appealing to emotions compared with hard propaganda (Hovland, Janis, and Kelley, 1953; Mattingly and Yao, 2022).

Additional evidence from literature beyond authoritarian propaganda also provides good reasons to believe in the persuasive potential of entertainment media. Firstly, entertainment products are known for their ability to attract attention. In the American context, it is well-documented that many voters regularly enjoy entertainment products (Baum, 2005; Baum and Jamison, 2006; Prior, 2005; Prior, 2005; Prior, 2007). In particular, Kim (2021) demonstrates how entertainment TV programs can effectively shape Americans' beliefs about economic mobility and their preferences for economic redistribution. Other studies examining access to liberal media in authoritarian regimes have similarly discovered the captivating influence that entertainment content holds over audiences (Chen and Yang, 2019; Kern and Hainmueller, 2009). A final line of research where we can observe an opportunistic view of entertainment's potential as a persuasion tool concerns the field of Entertainment-Education, a domain where practitioners and scholars aim to create entertainment products to convey information, fostering policy awareness, and encouraging behavioral change (Green, 2021; Green, Wilke, and Cooper, 2020; Green, Groves, and Manda, 2020).

Still, existing evidence on the persuasive effectiveness of entertainment-based propaganda, particularly regarding core political attitudes, remains limited. For example, seminal work by Hovland, Janis, and Kelley (1953) revealed that films created to educate American soldiers during World War II conveyed significant factual information but largely failed to increase soldiers' willingness to serve. Similarly, Mattingly and Yao (2022) found in their study on nationalistic propaganda in China, including television dramas, that they did not alter perceptions of government performance. Additionally, a series of field experiments on Entertainment-Education showed that while entertainment products could engage the target population and facilitate learning, they may not always lead to transformative changes in core political attitudes (Green, 2021; Green, Wilke, and Cooper, 2020; Green, Groves, and Manda, 2020). However, the examined products may not be comparable to the most captivating entertainment products found in pop culture. Educators often prioritize accuracy over entertainment in public education efforts. For instance, researchers in Hovland, Janis, and Kelley (1953) expressed concern that movies produced were too "Hollywood" to convey a persuasive message effectively. Similarly, the television dramas examined in Mattingly and Yao (2022) lack

widespread recognition or popularity. These products may not be as appealing as genuinely popular entertainment products, such as those examined in the American context by Kim (2021).

Entertainment products are essential for dictators not only because they are targets for censorship, as suggested by Esberg (2020), but also because dictators have the capability to produce highly appealing entertainment products that ultimately serve the goals of propaganda. An notable example is China, a country with one of the most extensive propaganda apparatuses in the world.⁴ China’s emphasis on soft propaganda is highlighted by its slogan “telling China’s story well,” signifying its aim to construct a favorable narrative about the regime and persuade individuals to embrace it. Notably, China’s propaganda blockbusters gained widespread popularity in the mid-to-late 2010s.⁵ Scrutinizing these celebrated productions allows for a reassessment of entertainment’s potential as a medium for disseminating authoritarian messaging, thereby shedding light on authoritarian regimes’ ability to shape public opinion.

I used *Wolf Warrior*, a 2015 action thriller with a patriotic element, as the case study for popular propaganda movies.⁶ This movie is about a soldier in the special forces conducting dangerous overseas operations. *Wolf Warrior* and its sequel have been compared to the Hollywood blockbuster *Rambo: First Blood*. In a controlled experiment, viewer compliance can pose a hurdle if the propaganda movies fail to captivate the viewers—yet the high entertainment value of *Wolf Warrior* enhances the feasibility of this experimental design. Nonetheless, the film’s propaganda message is far from obscure. Even though the movie was primarily produced by private companies, the Television and Art Center at the Political Department, affiliated with the Nanjing Military Region, was a co-producer. Memorable lines include “Anyone who offends China will be punished, no matter the distance.” The climactic scenes, which feature a fight over a Chinese flag in the form of a cloth patch on the soldier’s arm, convey a clear patriotic message.⁷ Another reason for selecting this

⁴This is evidenced by scholarly works such as Brady (2009); Stockmann (2013); King, Pan, and Roberts (2014); Lorentzen (2014); Han (2015); Huang (2015); Huang (2018); Repnikova (2017); Esarey, Stockmann, and Zhang (2017); Roberts (2018); Pan, Shao, and Xu (2020); Mattingly and Yao (2022)

⁵A concurrent paper explores how regimes organize propaganda production in the movie industry and the popularity of these movies, including China’s use of incentives to encourage private sector filmmakers and resources to produce propaganda.

⁶To approach this question, there are three approaches: evaluating the effects of a single film, assessing the effects of a category of films, and manipulating elements within a film to isolate potent components (Hovland, Janis, and Kelley, 1953). Since the subjects of interest have received little prior attention, it is essential to first understand the overall impact of propaganda movies as bundled treatments. Given time and budget constraints, starting with a single film rather than a category is a practical choice.

⁷While more research is needed to provide a definitive answer regarding whether the effects of this film are generalizable to the effects of other propaganda films, a parallel project of mine, which explores perceptions of

film as the treatment is its significance within contemporary China. The sequel, *Wolf Warrior II*, became the highest-grossing film upon its premiere. This series gained such immense popularity that the term “Wolf Warrior” was adopted as a nickname for China’s increasingly assertive foreign policy, underscoring the substantial cultural impact of these movies.⁸

Three categories of effects came under scrutiny in this research: 1) audience attention, emotional reaction, and evaluation; 2) changes in political opinions; and 3) the change in media consumption behaviors. Furthermore, I addressed the issue of long-term changes in media consumption behaviors by examining viewers’ subsequent viewing choices. A common concern in experimental designs like mine is whether the provided information can induce a substantial and lasting shift in people’s opinions, particularly in our information-rich world. While it is logistically challenging to re-survey participants after a month while minimizing attrition, I sought to determine whether respondents would voluntarily choose to engage with similar content over the course of a month. This can be a crucial prerequisite for a piece of propaganda to generate enduring persuasive effects.

Yet another concern with the persuasive power of propaganda is whether they are “preaching to the choir”. Many scholars argue that propaganda does not necessarily convert individuals with opposing beliefs, but rather, reinforces the beliefs of a likeminded audience. According to Welch (2008), a historian on Nazi propaganda, effective propaganda “must in a sense preach to those who are already partially converted” (p. 9). Several quantitative studies examining the effects of propaganda within the context of the Nazi regime and beyond have provided empirical evidence supporting the heterogeneous treatment effects of propaganda (Adena et al., 2015; Voigtländer and Voth, 2015; Peisakhin and Rozenas, 2018). Therefore, when designing research to study the effects of propaganda, it is crucial to consider the role of preexisting beliefs. One approach to address this issue involves classifying participants based on their preexisting preferences for propaganda or their preexisting attitudes toward the regime. This classification enables us to analyze how propaganda resonates differently with audiences who already hold varying views.

propaganda films, indicates that this movie is considered more entertaining and less indoctrinating than most other Chinese propaganda films. the entertainment element amplifies the persuasive power of propaganda, it is likely that my chosen film, *Wolf Warrior*, yields greater persuasive effects compared to other propaganda movies.

⁸The first movie, while still commercially successful, was not as sensational, which was crucial for my experimental design. To maintain balance, the experimental design excluded participants who had seen any of the movies used in the experiment. The fact that fewer individuals had viewed the first *Wolf Warrior* film provided more room for studying the treatment effect of this movie among an audience first exposed to it.

3 Experimental Design

3.1 Experimental Procedure

In my month-long experiment, I provided participants with accounts for a major movie streaming platform in China offering one-month subscriptions. The streaming platform is IQiYi, the most widely used streaming platform in the country. I also randomly assigned participants to one of three treatment groups with equal probabilities, where participants were encouraged to start with different recommended movies but then to continue enjoying the one-month subscription as they pleased. I surveyed their opinions after watching the recommended movie and tracked their viewing history over the following month. The full experimental procedure is illustrated in the flowchart shown in Figure 1.

A detailed description of the experimental procedure is included in Appendix A.1.

3.2 Movie Selection

As previously discussed, I used *Wolf Warrior*, a 2015 action thriller with a patriotic element, as the treatment propaganda film. The placebo movie is *Legendary Assassin* (the literal translation of its Chinese name is “Wolf Teeth”), a 2008 Hong Kong action thriller about an assassin who killed an evil gangster and ran away. It was made by the same director as the *Wolf Warrior* series, Wu Jing, who was also the lead actor in both movies. The reason for choosing this film as the placebo movie is its similarity to the *Wolf Warrior* films: They share the same genre, director, and lead actor. They also share the wolf-related theme and a similar artistic style. However, unlike the *Wolf Warrior* series, which centers on a soldier from the special forces, *Legendary Assassin* features outlaws and was produced in Hong Kong; thus, it does not contain any political relevancy.⁹

One key distinction between the treatment and placebo movies lies in the inclusion of propaganda content. The placebo served mainly to ensure that individuals in the control group watched a movie similar to the treatment but without influencing their political opinions. However, I do not assert that there are no differences in entertainment value between these two films. Given the differing popularity levels of these movies, respondents who had previously viewed either film were screened

⁹One concern is that somehow the presence of this actor will change people’s political opinions, because the actor is widely associated with the *Wolf Warrior* series. This is unlikely, however—the movie should bring the audience into the specific character in the movie, the assassin, which is different than just presenting this actor.

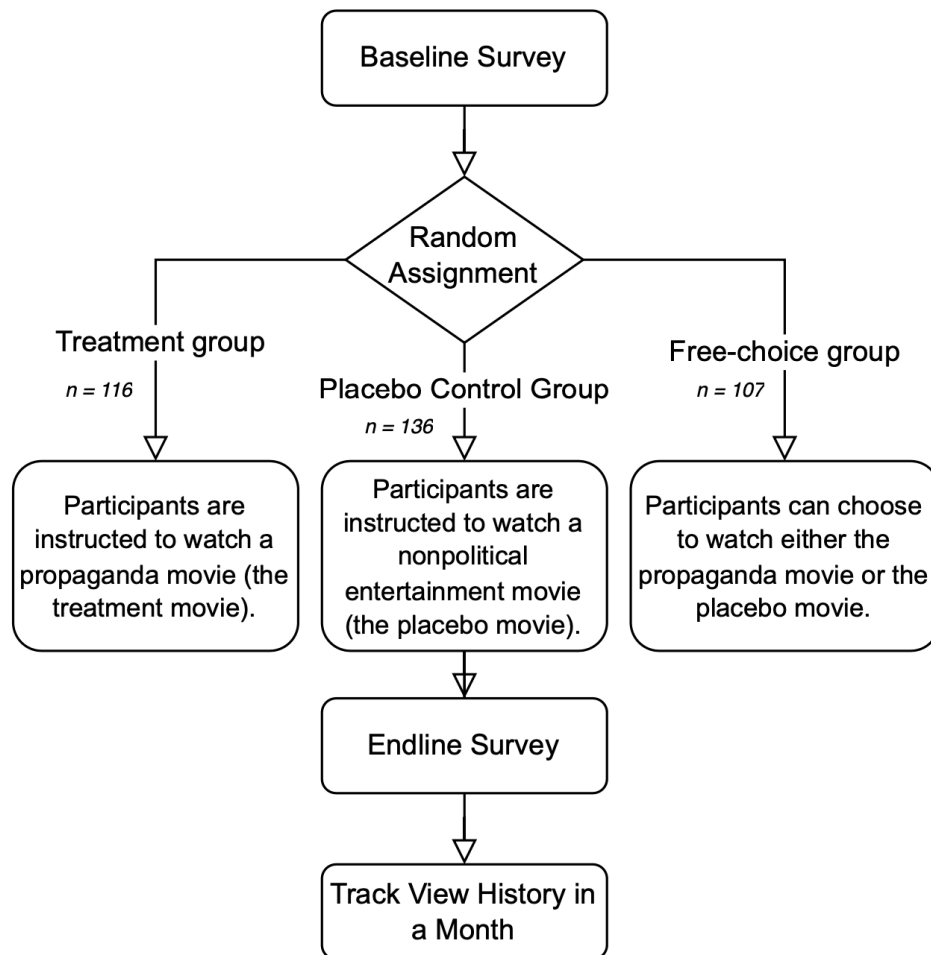


Figure 1: Experimental Procedure: Flowchart



Treatment: Wolf Warrior



Placebo: Legendary Assassin



Free-choice condition: Choose One Among the Two

Figure 2: Selection of Movies

out from participating to maintain balance.

3.3 Outcomes

I pre-registered a set of survey outcomes encompassing emotional responses to movies and opinion outcomes. Concerning opinion outcomes, I selected five main variables of interest: nationalism, economic perception, system pride, perceived government responsiveness, and willingness to protest. These variables were chosen based on a combination of factors, including prior literature, pilot surveys, and theoretical expectations, as detailed in the Appendix A.2. I also pre-registered behavioral outcomes, which include the movie participants selected to watch and the number of propaganda movies they watched in the following month.

For each of the five primary opinion outcomes, an index was calculated by averaging standardized responses from several questions to minimize measurement errors. These questions were intentionally designed to be standard and commonly used but varied in format as much as possible to mitigate issues such as acquiescence bias. In Appendix A.3, I have included the correlation matrices for individual question responses that measure the same outcome. The question details will be introduced when discussing the relevant results.

3.4 Estimand and Estimation

In this paper, we will mainly focus on the intent-to-treat effects obtained by comparing the outcomes of all subjects assigned to different treatment groups because the compliance with the treatment was expected to be high.

My empirical analysis is based on estimating regressions of the following form:

$$Y_i = \alpha + \beta_1 T1_i + \beta_2 T2_i + \beta_3 \mathbf{X}_i + \epsilon_i$$

Y_i refers to the outcomes. $T1$ refers to whether an individual is assigned to the forced-exposure group. $T2$ refers to whether an individual is assigned to the free-choice group. $\mathbf{X}_i$ represents the control variable(s). ϵ_i is the error term.

I reported the regression results from two models: a simple regression with only a treatment indicator, omitting covariates; and a regression model that also includes demographic variables as covariates to be adjusted.

Given the patient preference trial design, I am equipped to estimate the treatment effects among those who are prone to watch propaganda movies versus those who are not. To accomplish this goal, I will employ an instrumental variable strategy. Compared to the placebo control group, in the free-choice group, some individuals were experimentally induced to choose to watch propaganda movies. I can estimate the proportion of compliers (also referred to as self-selectors, those who would choose to watch the propaganda movie if assigned to a free-choice group) and the treatment effects specific to them using the instrumental variable approach. Similarly, compared to the treatment group, in the free-choice group, some individuals were experimentally induced to choose *not* to watch propaganda movies. Thus, I can estimate the proportion of non-selectors (those who would choose not to watch the propaganda movie if assigned to a free-choice group) and the treatment effects among them using the same instrumental variable approach.

However, it is important to note that my pilot survey suggests that most respondents may opt for the propaganda movie, resulting in insufficient statistical power for this particular estimation. Therefore, this exercise should be considered exploratory, as stated in my pre-analysis plan, which was pre-registered.

3.5 Participants and Subgroups

The participant enrollment took place from February 24, 2023, to March 31, 2023. Endline surveys were collected between February 25, 2023, and April 6, 2023. A total of 362 participants were enrolled in the study. Among them, 116 participants were assigned to the treatment group, 136 to the placebo control group, and 107 to the free-choice group. The final survey included responses from 101 participants in the treatment group, 116 in the placebo control group, and 93 in the free-choice group. For additional details related to compliance and attrition, please refer to Section 4.3.1.

This study’s sample displays biases towards certain demographics, with over-representations of male, younger, and higher-educated respondents. The extent of this bias is represented in Appendix A.5, illustrating the distribution of gender, education, and age among the participants.

Preference for propaganda movies, which was identified in prior to the treatment delivery, is a crucial aspect of heterogeneity explored in this study. In the baseline survey, respondents were asked, “What types of movies do you usually prefer to watch? Please select all that apply.” They were provided with 12 options, ranging from romance and comedy to action and main melody movies, among others. Subsequently, they were asked, “What types of movies do you usually prefer not to watch? Please select all that apply.” The same 12 options were presented. Among the options, one of them would be referred to as the “main melody movie” (MM for short in this paper). The term “main melody movie”, in the Chinese context, refers to cultural products that actively promote the state ideology. If participants chose “main melody movie” for either of the questions, they were classified as liking or disliking propaganda movies, respectively.

The distribution of respondents’ taste for propaganda movies reveals that the majority of respondents were initially indifferent to this genre ($n = 223$). Among the rest, I observed that there were more respondents who had a distaste for propaganda movies ($n = 83$) than those who had a preference for them ($n = 51$). One respondent selected “Main Melody Movie” for both the “like” and “dislike” questions.

Due to the small sample size of the group that expressed a preference for propaganda movies, I collapsed this group with those who expressed indifference for the purpose of analysis, creating two subgroups based on whether respondents expressed a distaste for propaganda movies or not. Thus,

analysis of the heterogeneous treatment effects can be conducted by examining the treatment effects among two subgroups: one consisting of respondents who did not express a distaste for propaganda movies ($n = 274$), and the other comprising respondents who did express a distaste for such movies ($n = 84$). Appendix 3 illustrates that male participants and those who express low satisfaction with the country’s overall situation are more prone to disliking propaganda movies, though these relationships do not achieve statistical significance.

4 Experimental Results

In this section, I will present and discuss the empirical results. First, I will examine the differences between the treatment group and the control group, excluding the free-choice group. Then, I will introduce the findings from the free-choice group. Next, I will conduct additional tests related to threats to identification. Finally, I will discuss the interpretation of the experimental results.

4.1 Comparing the Treatment Group and Control Group

The first part of the results I present focuses on the differences between the treatment group and control group, which indicate the effect of the film on various dimensions. Alongside plotting estimates of treatment effects and confidence intervals, I also utilize an alternative style of barplots to present my results. In these plots, the bar labeled “Placebo Control” represents the mean value of the control group, while the “Treatment” bar indicates the mean value of the treatment group, equivalent to the control group plus the estimated treatment effect. Additionally, the error bar depicts the estimated confidence interval of the treatment effect, equivalent to the control group’s mean value plus the lower bound to the upper bound of this 95% confidence interval. This barplot format enhances visibility of the treatment effect’s magnitude relative to the absolute control group mean.

4.1.1 Emotions

Results from the experiment suggests that the propaganda movie is quite effective at generating interest and capturing attention. Among participants who registered for the study, 12.26% of participants in the treatment group and 14.18% of participants in the control group did not show

signs of using the provided account. Another 11.32% of participants in the treatment group and 11.94% of participants chose to use the account without clicking on the recommended movie. Among the remaining participants who did start watching the movies, 87.65% of participants in the treatment group and 90.91% of participants in the control group watched more than 90% of the recommended movies.

Participants also rated the propaganda movie quite favorably. In the post-viewing survey, participants were asked to rate the movie they watched on a scale from 0 to 10. As shown in Figure A.7 in the Appendix, the mean rating for the placebo movie was 7.59, while for the propaganda movie, it was 8.54. Even among those who previously disliked propaganda, the treatment film was not viewed less favorably than the placebo. Further discussion about attrition and compliance, including additional details and alternative measures of compliance, will be provided in a later section titled “Threats to Identification.”

In this survey, I also asked questions that assessed the emotional experiences of viewers while watching the movie. The survey presented four distinct emotional states—sadness, happiness, anger, and pride—following the practice of Mattingly and Yao (2022). Respondents were asked to rate their emotional experiences using a five-point scale, including the options of “almost none,” “relatively few,” “moderate,” “relatively many,” and “extremely many.” Participants’ responses are presented in Figure 3.

We can see that, as the baseline, participants watching the placebo control movie reported mild emotional experiences while watching the movie, consistent across different emotions. In contrast, participants watching the treatment movie reported mildly stronger feelings of happiness and sadness, and substantially stronger feelings of anger and pride. Most notably, participants reported an average feeling of pride above the second-highest level, “relatively many.” This suggests that this particular propaganda movie elicited especially powerful feelings of pride.

A notable pattern emerges when participants are grouped based on their prior attitudes toward propaganda: they generally exhibit similar emotional responses during movie-watching, particularly in terms of sadness, happiness, and anger levels. The most notable difference arises in their experience of pride. While participants without a prior distaste for propaganda display exceptionally high levels of pride when exposed to this piece of propaganda, those with a prior distaste only experience a relatively mild increase in pride. Nevertheless, even among participants who previously

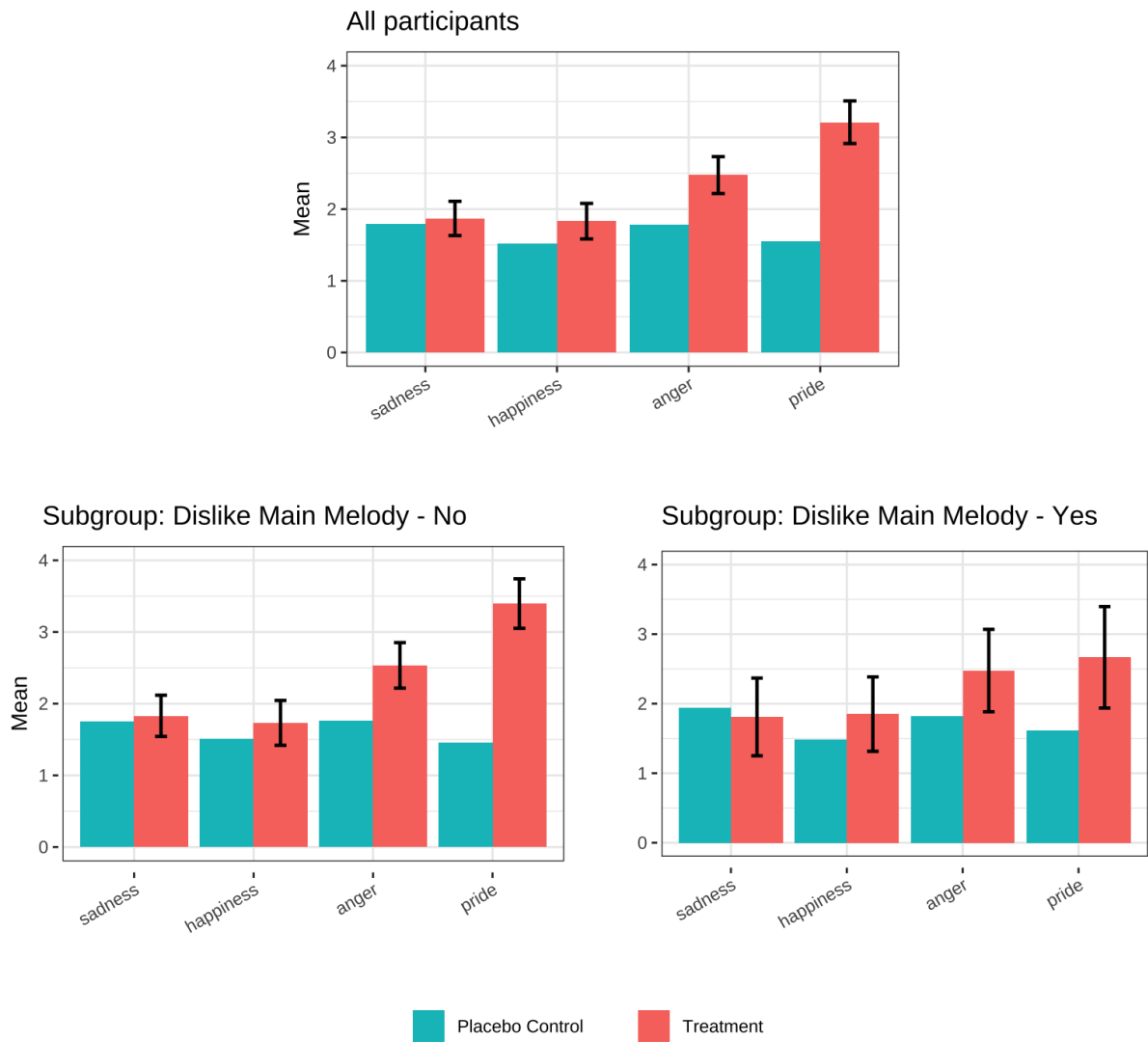


Figure 3: Emotions During Movie-Watching

disliked propaganda, pride remains the strongest emotional reaction when exposed to this particular propaganda movie. This heightened level of pride significantly surpasses their emotional response to a placebo control movie. Therefore, to a certain extent, the chosen treatment movie can override preconceived aversions to propaganda, which indicates the potent influence of popular propaganda content on eliciting emotional responses across various audiences.

4.1.2 Changes in Opinions

In Table 1, I present the results of linear regression models for five outcome variables, including nationalism, economic perception, system pride, perceived responsiveness, and willingness to protest, where the predictor variable is the treatment status (control or treatment). The main outcomes are obtained by averaging standardized responses to various questions. The details of the questions and the results for each question will be presented shortly.

The results in “Panel 1: All Participants” suggest that the propaganda film had a possible effect on enhancing nationalism with an effect size of 0.114, although this effect is not statistically significant, given the standard error of 0.123. The film positively influenced people’s economic perceptions, with an effect size of 0.175, and this result is significant at 10% level. The propaganda film increased pride in the political system with an effect size of 0.137, but this is not statistically significant. The film increased perceived responsiveness with an effect size of 0.166, a result that is significant at 10% level. Lastly, it had a slight dampening effect on the willingness to protest, with an effect size of 0.055, though this is not statistically significant.

To further analyze the data, I investigate subgroup effects based on participants’ pre-treatment expressions of dislike for propaganda. The results are presented in “Panel 2: Dislike MM – No,” and “Panel 3: Dislike MM – Yes.” In “Panel 2: Dislike MM – No,” the estimated treatment effects on those who did not express prior distaste for propaganda were consistently positive and statistically significant across all outcomes, except for willingness to protest. Among this group of subjects, the propaganda film substantially increased outcome values including nationalism, economic perceptions, system pride, and perceived responsiveness by approximately a 0.3 standard deviation.

However, in “Panel 3: Dislike MM – Yes,” the treatment yielded negative effects on all outcome variables except for the willingness to protest. Notably, these negative effects were more pronounced compared to those observed in individuals with a favorable view of propaganda. The most substantial

Table 1: Main Results: Effects of Propaganda Movie on Opinions

	Nationalism	Economic Perception	System Pride	Perceived Responsiveness	Willingness to Protest
(Intercept)	-0.052 (0.084)	-0.107 (0.065)	-0.052 (0.071)	-0.053 (0.066)	-0.062 (0.070)
Treatment	0.114 (0.123)	0.175+ (0.096)	0.137 (0.105)	0.166+ (0.098)	0.055 (0.104)
Num.Obs.	217	219	219	219	219
<i>Panel 2: Dislike MM - No</i>					
(Intercept)	-0.008 (0.091)	-0.088 (0.070)	-0.034 (0.078)	-0.027 (0.071)	-0.027 (0.090)
Treatment	0.272* (0.131)	0.319** (0.100)	0.291* (0.112)	0.322** (0.101)	0.063 (0.130)
Num.Obs.	157	159	159	159	159
<i>Panel 3: Dislike MM - Yes</i>					
(Intercept)	-0.149 (0.167)	-0.150 (0.133)	-0.094 (0.142)	-0.113 (0.134)	-0.142 (0.099)
Treatment	-0.435 (0.265)	-0.303 (0.211)	-0.374 (0.224)	-0.359+ (0.211)	-0.002 (0.157)
Num.Obs.	60	60	60	60	60

negative effect was observed on nationalism, where the treatment shifted outcome values by less than half a standard deviation. For other variables, the treatment produced a shift of more than 0.3 standard deviations. This suggests that the treatment may have had an inverse effect on participants who initially disliked the target of the propaganda, although these results are not statistically significant.

Figure 4 presents a visual representation of the estimated subgroup effects. Compared to the other group, participants who had previously expressed distaste for propaganda movies exhibited slightly larger absolute estimated effects, although these effects are less precisely estimated. This does not necessarily imply that this group had an opposing emotional experience during their viewing of the movie, as both groups showed a roughly similar emotional pattern. Instead, it seems more plausible that this group disagreed with the movie's message and subsequently adjusted their beliefs in the opposite direction.

This variation in treatment effect between the two subgroups could lead to the overall effect of the treatment, when averaged across all participants, appearing statistically insignificant. In other

words, the positive effect of the treatment on the first subgroup—those who did not express a prior distaste for propaganda—could be partially “cancelled out” by the negative effect on the second subgroup, those who expressed a distaste for propaganda movies, resulting in an overall effect that is small or even negligible.

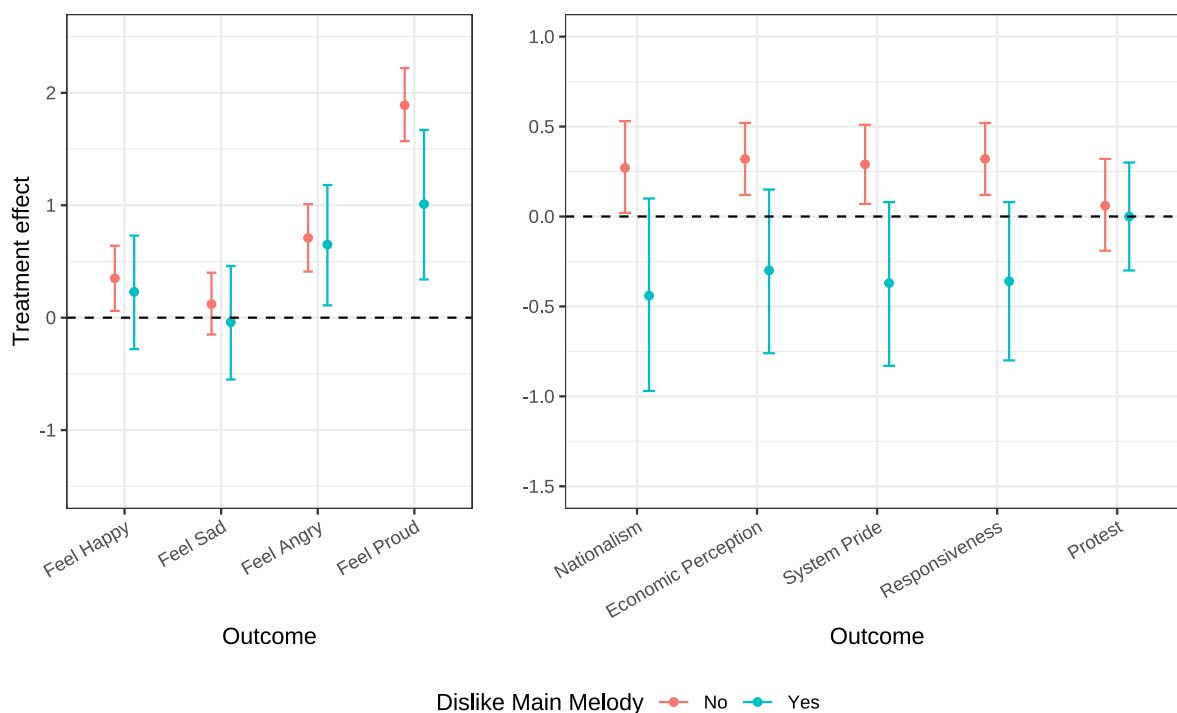


Figure 4: Treatment effects by outcome and taste for propaganda

Let us now consider the pattern based on the original questions and scales used. Figure 5 displays the individual questions that comprise the nationalism index. The respondents initially exhibited a high level of nationalism. However, those who did not express dislike for main melody movies still experienced a small increase in their sense of nationalism after watching the propaganda movies, further elevating the already high baseline level of nationalism. Conversely, those who expressed prior distaste for propaganda movies tended to react differently, displaying negative treatment effects. It is worth noting that certain questions—particularly those related to extreme forms of nationalism (such as “nationalism1” and “nationalism2”)—elicited greater backlash compared to milder questions about national pride (such as “national_pride1” and “national_pride2”). While the treatment had only a marginal effect on the participants’ responses to milder questions, it significantly influenced their responses to more extreme questions, as well as their willingness to

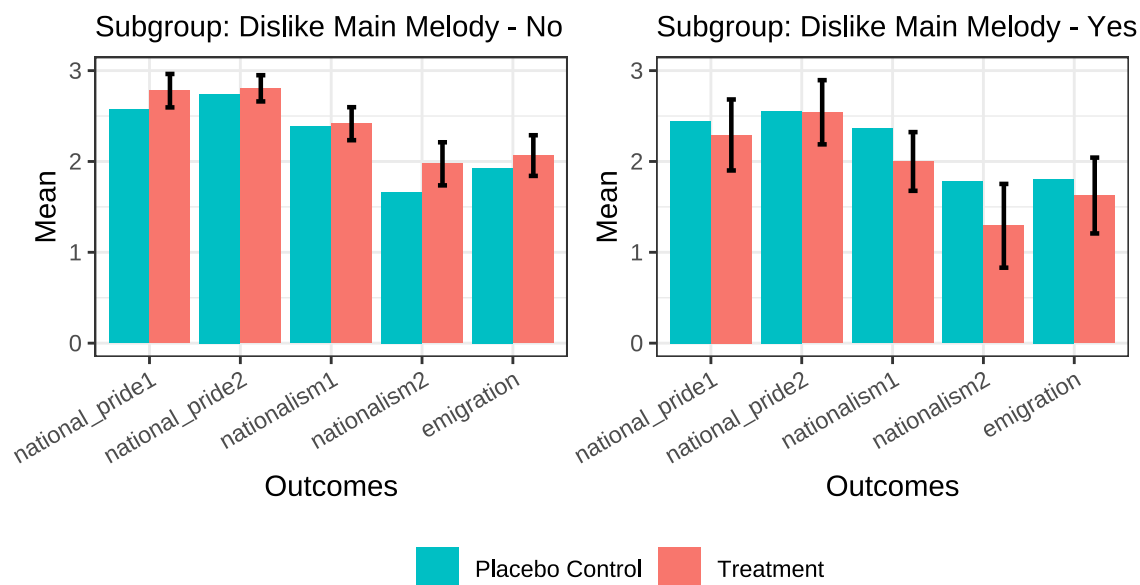
emigrate, with a notable shift of 0.5 on a 5-point Likert scale.

Figure 6 presents the individual questions that pertain to economic perception. Participants who did not express dislike for main melody movies experienced an increase in their economic perception after watching the propaganda movies, meaning their confidence in themselves and the country's current economic situation experienced a substantial boost, with an increase of 0.5 on a 5-point Likert scale. However, the outlook for the future, which was initially more positive, only saw a marginal increase. On the other hand, those who expressed a prior distaste for propaganda movies only experienced an increase in the perception of their current economic situation, at a size comparable to that of the other group. While they had a slightly lower outlook for their own economic situation in the future, this group significantly downgraded their outlook for the country's economic situation. This downgrade had a notable effect on both their present and future perceptions, resulting in a shift of approximately 0.5 on a 5-point Likert scale.

In regard to system pride, as demonstrated in Figure 7, the individual questions reinforce the observation of a divergent response to the treatment. A majority of participants who did not express distaste for main melody movies experienced a steady increase in their system pride after viewing the propaganda movies, equaling roughly 0.2 on a 5-point Likert scale, as shown across responses to questions. The remaining participants experienced a decrease in their sense of national pride. The most significant disagreement was observed in response to the statement that the political system deserves people's blind support even if it encounters problems, resulting in a downgrade of approximately 0.5 on a 4-point Likert scale. Moreover, participants exhibited an increased belief that the regime requires change, with an average shift of 0.2 on a 3-point Likert scale. However, they only marginally lowered their confidence in the regime's ability to solve problems.

Regarding perceived responsiveness, as illustrated in Figure 8, the individual questions once again affirm the existence of a varied response to the treatment. In particular, the majority of participants who did not harbor a prior dislike for main melody movies experienced an elevation in their perception of government responsiveness after viewing the propaganda movies. Although the increase was modest, it demonstrates a consistent pattern across questions, amounting to an average increase of approximately 0.2 on a 4-point scale. Conversely, the remaining participants experienced a decrease of a similar magnitude in their perception of government responsiveness.

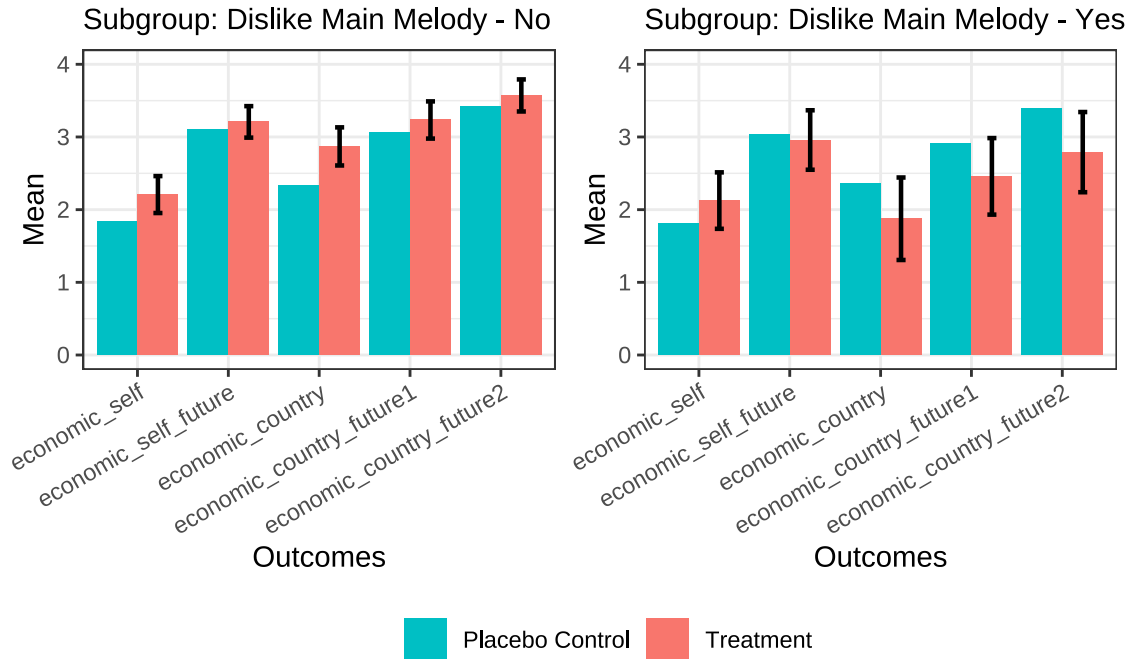
Finally, as depicted in Figure 9, propaganda movies did not reduce participants' willingness to



Index	Questions	Scales
1 national_pride1	As a Chinese, I feel very proud.	0 = Strongly disagree, 3 = Strongly agree
2 national_pride2	I am proud of China's history and arts.	0 = Strongly disagree, 3 = Strongly agree
3 nationalism1	Generally speaking, China is better than most other countries.	0 = Strongly disagree, 3 = Strongly agree
4 nationalism2	People should support their country even if the country is wrong.	0 = Strongly disagree, 3 = Strongly agree
5 emigration	Given the chance, how willing would you be to live in another country?	0 = Strongly willing, 3 = Strongly unwilling

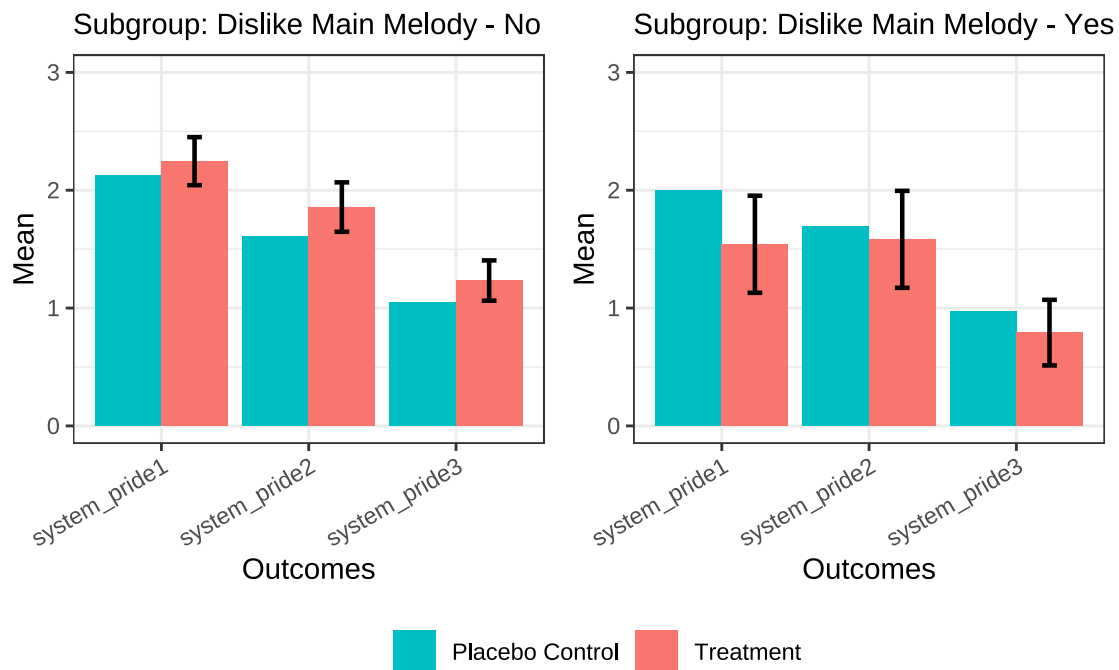
Figure 5: Nationalism: Individual Questions

Note: The bar labeled “Placebo Control” represents the mean value for the control group, while the “Treatment” bar denotes the mean value of the control group plus the estimated treatment effect. Furthermore, the error bars represent the range from the control group’s mean value plus the lower limit to the upper limit of the 95% confidence interval for the estimated treatment effect.



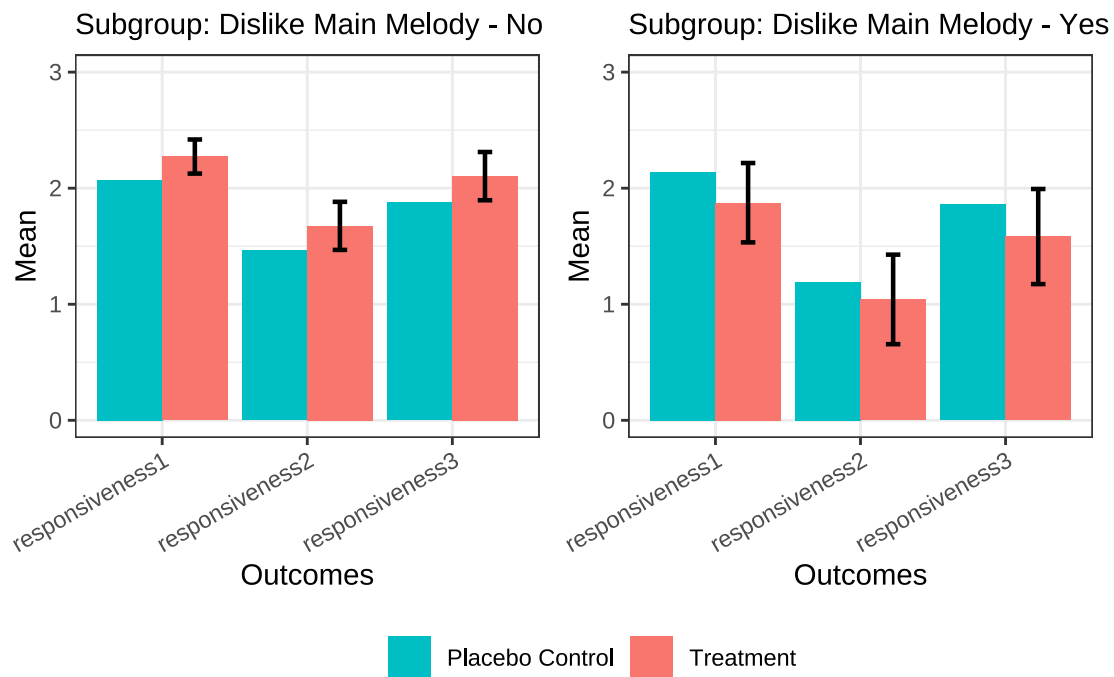
	Index	Questions	Scales
1	economic_self	Overall, how would you rate your current economic situation?	0 = Very bad, 4 = Very good
2	economic_self_future	Overall, how would you predice the change of your economic situation in five years?	0 = Much worse, 4 = Much better
3	economic_country	Overall, how would you rate the country's current economic situation?	0 = Very bad, 4 = Very good
4	economic_country_future1	Overall, how would you predice the change of country's current economic situation in a few years?	0 = Much worse, 4 = Much better
5	economic_country_future2	How do you think your child's standard of living will compare to yours when they are your age (if you don't have a child now, assume you have)?	0 = Much worse, 4 = Much better

Figure 6: Economic Perception: Individual Questions



Index	Questions	Scales
1 system_pride1	A political system like ours, even if it runs into problems, deserves the people' s support.	0 = Strongly disagree, 3 = Strongly agree
2 system_pride2	Our political system is capable ofn solving the most pressing problems our societyn faces.	0 = Strongly disagree, 3 = Strongly agree
3 system_pride3	Would you say our system of governmentn works fine as it is, needs minorn change, or needs major change?	0 = It needs major change, 2 = It works fine and no need for changes

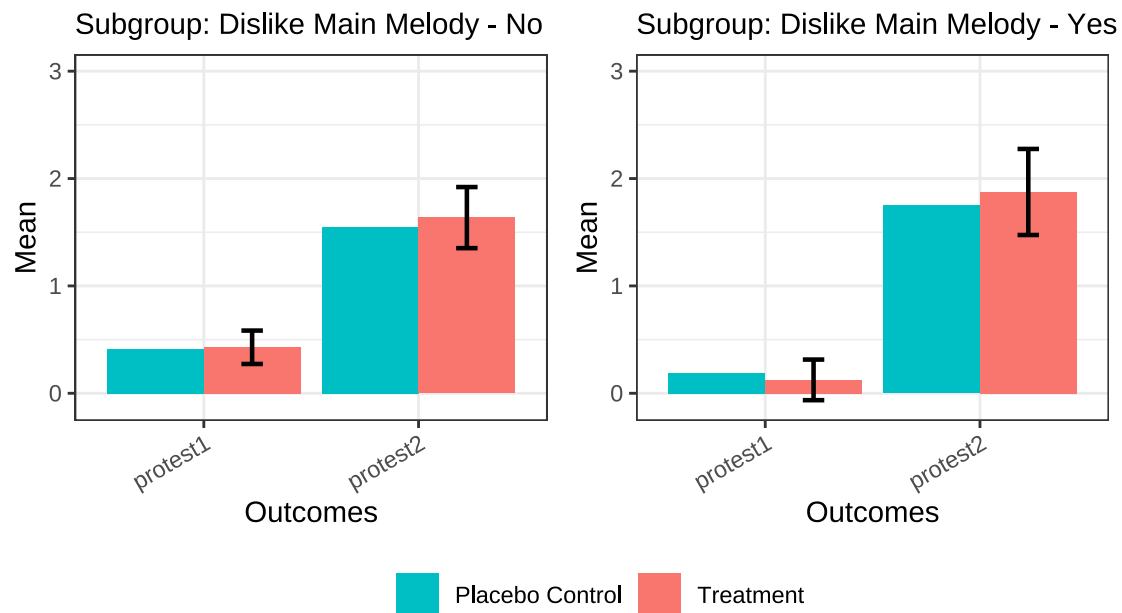
Figure 7: System Pride: Individual Questions



Index	Questions	Scales
1 responsiveness1	Overall, our government is working for the people and responsive to the needs of people.	0 = Strongly disagree, 3 = Strongly agree
2 responsiveness2	Currently, there are not enough legal channels for ordinary people to influence government behaviors.	0 = Strongly disagree, 3 = Strongly agree
3 responsiveness3	How responsive do you think the current government is to people's needs?	0 = Very unresponsive, 3 = Very responsive

Figure 8: Perceived Responsiveness: Individual Questions

protest. The study design is unable to produce an explanation of why this should be the case, but this could potentially be attributed to the dual-edged nature of propaganda films. While they may heighten people’s satisfaction with the regime, consequently diminishing their inclination to protest, the intense emotional responses they incite could also make people more agitated and, thus, more inclined to protest.



Index	Questions	Scales
1 protest1	If you have encountered very difficult situations or been treated very unfairly in your life, would you consider seeking solutions through the following channels? (choose all that apply)	0 = "Offline petitions and protests" is not chosen, 1 = "Offline petitions and protests" is chosen
2 protest2	If you and a group of people experienced very unfair treatment in your life, and someone organized a gathering or group walk, would you consider participating?	0 = Definitely no, 3 = Definitely yes

Figure 9: Willingness to Protest: Individual Questions

4.1.3 Consumption of Propaganda Movies in the Subsequent Month

As I examined people’s consumption of propaganda movies in the following month, depicted in Figure 10, there was no compelling evidence that the initial propaganda film sparked much additional interest in such content. Even without any intervention, respondents who initially held no distaste for propaganda films already showed a higher baseline for viewing such content than those who did. However, this count appears to grow only slightly due to the treatment. Conversely, respondents who initially reported disliking propaganda did show a modest rise from their considerably low baseline viewing habits. Both increases are not statistically significant.

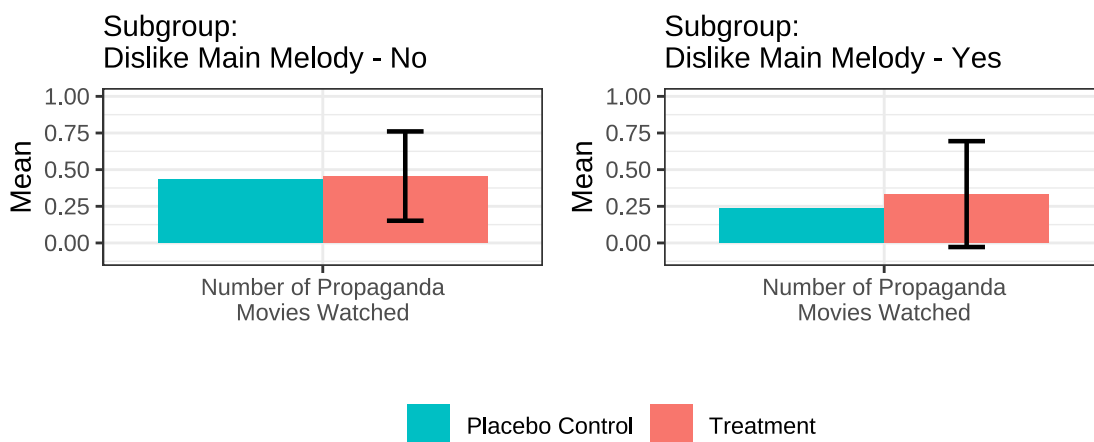


Figure 10: Actual number of propaganda movie watched following the treatment

These behavioral findings present an interesting contradiction to participants’ stated preferences. As shown in Figure 20 in the Appendix, when asked about their willingness to view a sequel to the propaganda movie in the endline survey, respondents who did not express a distaste for propaganda films reported a heightened interest in viewing similar content as the result of the treatment. In stark contrast, those who had initially expressed distaste for propaganda report a decreased willingness to watch the sequel as the result of the treatment.

These findings suggest that while the treatment seems to polarize the existing sentiments toward watching propaganda films, it does not effectively change the actual consumption behavior in the subsequent period. People’s unwillingness to act on their expressed taste suggests that many more factors affect people’s consumption choices related to political content aside from their political

preferences.

The behavioral data further emphasizes the natural diversity of people’s entertainment preferences. The movie website offers a wide range of options, making propaganda movies a relatively insignificant genre in the vast sea of entertainment. On average, participants watched 8.56 movies using their account, with the maximum number reaching 85. In contrast, the average number of subsequent propaganda movies viewed following the assigned film was 0.47, and the highest number observed so far was 6. Among all propaganda movies watched by more than 1%, the average viewing percentage was 58.43%.

The pie charts in Figure 11 illustrate the various types of entertainment products that people clicked on in the subsequent month. It shows that non-propaganda dramas dominated entertainment consumption, accounting for 62% of the total consumption. Propaganda dramas accounted for a smaller 6.88% of the entertainment pie, suggesting a general inclination among viewers towards narratives that steer away from explicit political or ideological content. This is much less than their current market share of around 20% in the box office.. The data also highlight the diversity in the audience’s taste, with considerable consumption not only of American and Hong Kong dramas but also of animations, variety shows, and other genres. In Figure 21 in the Appendix, I plot the entertainment products clicked on by treatment groups, and we can see that all three groups exhibited a roughly similar pattern of diverse entertainment taste.

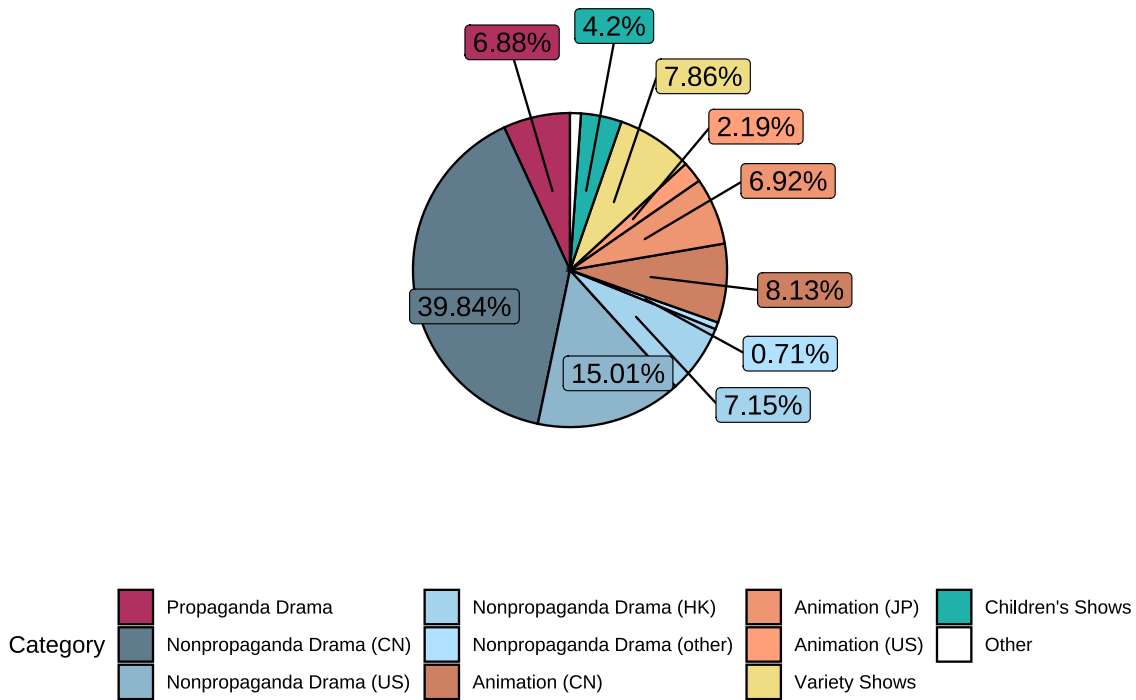


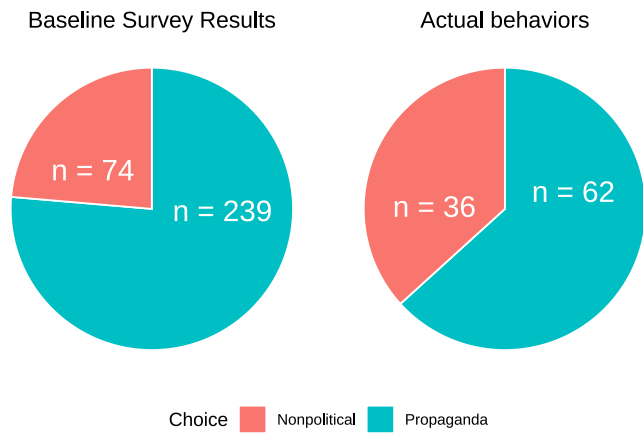
Figure 11: Audience's Taste in Entertainment

4.2 Analyzing the Free-Choice Group

What do people choose when offered a choice between the treatment movie and the placebo control movie? In the baseline survey, respondents were asked to indicate their preference between the treatment movie (propaganda) and the placebo control movie (non-propaganda). To mask the actual treatment, three pairs of movies were presented for the respondents to choose from. As shown in Figure 12, in the survey, 76.36% of respondents indicate a preference for the propaganda movie.

However, when participants in the actual free-choice group decided which movie to devote 90 minutes to, only 63.27% of them chose the treatment movie. This number is smaller than what they indicated in the survey results, indicating a gap between what people say and what they actually do. The table confirms that more people switched from propaganda movies to non-propaganda movies than the other way around. These results show that while the propaganda movie is the more popular choice in both the survey and the actual free-choice group, there is a discrepancy between the stated preferences and actual choices.

Appendix A.10 shows that male participants, younger participants, those with higher education



		Expressed preference	
		Propaganda	Nonpolitical Entertainment
Actual choice	Propaganda	54	8
	Nonpolitical Entertainment	14	22

Figure 12: Expressed preference vs actual choice

levels, individuals from lower-tier cities, and those with lower incomes were more likely to opt for the propaganda movie instead of the nonpolitical movie. However, none of these differences are statistically significant.

Tables 6 to 9 in the Appendix present the mean and standard deviation of the outcome values from different groups. Consistent with common expectations, the mean outcome in the free-choice group generally falls somewhere between the placebo group and the treatment group. This suggests that if we divide respondents into those who would choose to consume propaganda given another choice and those who would not, propaganda is unlikely to have a negative average treatment effect on either group.

Appendix A.11 provides the intent-to-treat (ITT) estimates for the average treatment effect across different treatment statuses. Appendix A.12 presents the instrumental variable (IV) estimates of the average treatment effect among self-selectors (those who would choose to watch propaganda films given an alternative choice) and non-selectors (those who would not choose to watch propaganda films given an alternative choice). It is important to note that this is an exploratory exercise with limited statistical power. However, the results do not reveal a consistent pattern across these groups.

4.3 Threats to Identification

4.3.1 Compliance and Attrition

While there was a certain degree of attrition observed in each group concerning the opinion outcomes, the rates of attrition were reasonably balanced across the treatment groups. Specifically, 15 participants or 12.93% of the treatment group, 20 participants or 14.02% of the placebo control group, and 14 participants or 13.08% of the free-choice group did not complete the final survey. In general, there is little evidence to suggest that participants' willingness to partake in the experiment is likely related to treatment assignment.

Several alternative measures were used to assess participants' attention to the recommended movies. The first strategy was to trace people's viewing history and see if they watched the recommended movie. Figure 23 and Tabel 11 in Appendix A.13 demonstrates that there is no differential compliance rate between the treatment and control groups. Additionally, Figure 24 in Appendix A.13 provides the estimates of the Complier Average Causal Effects, the average treatment

effects among those who paid attention to the film. Here, it is measured as those who watched the movies above 90 percent. The results suggest that while the estimates vary slightly from the previous results, the pattern remains fairly consistent.

As another strategy to measure compliance, I assessed participants' ability to identify the correct elements, plots, and lines from a list of options in the final survey. Among participants who completed this survey, 91.44% achieved an accuracy rate of above 60%, while 80.82% achieved an accuracy rate of above 80%. Furthermore, 42.46% achieved a perfect accuracy rate of 100%. These findings confirm that most participants complied with their assigned treatment. The Complier Average Causal Effects are estimated again, using an instrumental variable approach where the attention score served as the instrument. As shown in Figure 25 in Appendix A.13, using this approach yields an overall pattern that remains consistent.

4.3.2 Manipulation Checks

In the final post-viewing survey, I asked participants to write a few sentences commenting on the movie they had watched. To discern whether participants had absorbed the propaganda messages from the movies, I analyzed the word clouds generated from the audience's comments. Figure 13 presents the results. The prominent words for the non-propaganda movie were "plot", "feeling", and "exciting". This suggests that viewers primarily focused on the storyline and the thrill of the film, characteristics typical of non-propaganda content. The absence of politically charged words or messages indicates that viewers were not impacted by any distinct ideological or political message from this movie.

In contrast, the comments for the propaganda film included words like "China," "patriotism," and "soldier." This implies that viewers noted the nationalistic and military-themed propaganda messages within the film, in addition to paying attention to the storyline. The presence of these words suggests that the film successfully conveyed its ideological content to its audience.

4.3.3 Balance and Covariate Adjustment

Due to the use of simple random assignment, covariate imbalances are observed across the groups, as presented in Appendix 12. Fortunately, after adjusting for these covariates, the overall results remain mostly consistent. These results are demonstrated in Appendix A.15. However, it is noteworthy

4.3.4 Other Heterogeneous Treatment Effects

Appendix A.17 illustrates that the treatment tends to negatively affect the opinions of those initially dissatisfied with the country’s overall situation, while having a positive effect on the rest of the participants. This confirms the previous finding that propaganda influences individuals with different political attitudes in distinct ways.

Appendix A.18 presents subgroup treatment effects among those who frequently consume movies and those who do not. In the real world, it can be reasonably deduced that participants who frequently consume movies are the ones most likely to be affected by the promotion of propaganda through movies. However, no clear distinction can be discerned between these groups in terms of how they are affected by propaganda movies.

Another concern arising from potential heterogeneous treatment effects by gender. In my experiment, a greater number of male participants registered, which could render the obtained estimates very unrepresentative if the true average treatment effects vary between genders. In Appendix A.19, I present the estimates for male and female participants separately to see if that is the case.

4.4 Discussion: Interpreting the Results and Their Implications

This research reveals that propaganda blockbusters can serve as a particularly well-received and uplifting medium for political messaging. This research shows that such popular propaganda elicits more positive than negative emotions, with pride being the most prevalent emotion. In contrast, the experiment that Mattingly and Yao (2022) conducted with an anti-Japanese TV drama stirs feelings of anger and sadness while suppressing a sense of happiness and pride. This anti-Japanese drama likely taps into the “Century of Humiliation” (1839–1949) narrative, a common thread in Chinese propaganda to fortify Chinese nationalism. However, the new commercial propaganda film primarily provokes positive emotions, mirroring China’s track as a rising power. While negative emotional experiences can inflame resentment toward a specific group, positive emotional experiences excel in fostering hope and shared purpose, and this research shows how they can also be of help to propaganda.

While creating popular propaganda films for authoritarian regimes poses challenges and requires

significant investment, my results suggest that the potential gains for the regime are considerable. Contrasting with Huang (2018)’ notion of “hard propaganda,” which erodes subjects’ opinions about the regime, the popular propagandist entertainment analyzed in this study notably enhances the audience’s perceptions of the regime. Unlike the propaganda discussed in Mattingly and Yao (2022), the material examined here has the capacity to influence opinions about the regime itself. The magnitude of the treatment effect among individuals who did not exhibit a preexisting aversion to propaganda is significant—comparable to the effect of increasing general satisfaction by one unit.¹⁰ This implies that authoritarian regimes may utilize propaganda as a means to shape political opinions rather than addressing underlying socio-economic conditions within the country. This movie’s ability to capture the hearts and minds of the audience is significant, especially given its extensive reach: *Wolf Warrior* attracted over 20 million viewers, while its sequel, *Wolf Warrior 2*, reached a staggering 100 million viewers. This underscores the pivotal role these movies play in shaping public sentiment and highlights their significance for the regime.

One of the most pronounced effects of this propaganda blockbuster is to foster a sense of optimism about current and future economic conditions. Despite prevalent literature suggesting the difficulty in swaying economic views (Frye, 2022; Coyne and Hall, 2021; Sobolev, 2019), this research indicates that while people’s economic understanding is rooted in personal experience, instilling hope can significantly reshape these perceptions. When propaganda is executed well, efforts to shape economic views could be a less contentious method of political messaging, making economic perceptions more malleable than political attitudes. This implies that authoritarian regimes do not necessarily suffer from a loss of support even during economic downturns if they use propaganda methods effectively to preserve a positive economic outlook among their people.

Despite these findings attesting to the effectiveness of entertainment-based propaganda, the propaganda method itself is not without limitations. For one, it does not necessarily cultivate a taste for similar content. This suggests that even if the propaganda blockbuster is initially popular, in order for such content to be continuously consumed by viewers, ongoing promotion and support from the regime might be needed. Building on my concurrent project about the production of entertainment-based propaganda, this implies that the regime must continually put in efforts, such

¹⁰In the baseline survey, individuals could choose from five options to express their overall satisfaction with China’s current economic and social conditions. The coefficients obtained when using this covariate to explain the outcomes are comparable to the estimated treatment effects.

as shaping the market to favor such movies over non-political entertainment. This has significant economic consequences, as government intervention of this nature can lead to substantial economic inefficiencies. In addition, we do not know how long-lasting the initial persuasive effects are, which might add to the need for the authoritarian regime to promote similar content continually to maintain support through such means.

For another, a small subset of viewers harbored worsened attitudes after watching the film. It suggests that, for authoritarian regimes, ill-targeted propaganda might as well boomerang, deteriorating people’s perception of the regime. Importantly, the decision to consume a movie depends on factors unrelated to political opinions, such as the movie’s brand name and its significance in popular culture. While this may be advantageous for the authoritarian regime in terms of attracting a wide audience and fostering a pro-regime popular culture, it could also result in the movie drawing viewers who are actually repelled by the political messaging, potentially leading to backlash after viewing it. Nonetheless, the fact that propaganda films positively influenced nearly 80% of the respondents is notable. After all, it remains unclear whether an authoritarian regime is truly concerned with garnering support from every single one of its citizens.

5 Conclusion

In 2021, Wang Xiaohui, the head of the Chinese Film Bureau at that time, proudly acclaimed how their work had contributed to the establishment of a strong atmosphere of love for the party, country, and socialism throughout the entire party and society (Wang, 2021b). This article argues that there is some element of truth to this statement, suggesting that entertainment-based propaganda, such as propaganda blockbusters, can indeed influence opinions in favor of the regime among the majority of the population. While the existing literature on authoritarian propaganda primarily centers on news media, this article fills a gap by providing evidence of the effectiveness of entertainment-based propaganda, offering a more comprehensive perspective on how authoritarian regimes shape public opinion.

On the other hand, this experiment also demonstrates that there are key limitations to the use of popular entertainment as propaganda. First, the intended effects were only observed among respondents who did not express a preexisting aversion to propaganda movies, accounting for

approximately 80% of all respondents. Among the remaining 20%, the estimated treatment effect might be in the opposite direction, although the estimation is less precise. Second, despite the positive reception of the propaganda movie, behavioral data indicate that people do not subsequently consume more propaganda movies. The experimental site, an actual movie streaming website, offers a large variety of entertainment options, including content that cannot appear on the big screen. Behavioral data suggests that people's preferences for entertainment are naturally diverse, and while there has been some voluntary consumption of popular propagandist entertainment products, it remains a relatively minor category compared to the wide array of diverse entertainment options available.

The current paper does not fully determine if entertainment products effectively persuade audiences to embrace all kinds of political messages. While the selected propaganda movie mainly features nationalistic themes, it is important to understand that Chinese propaganda encompasses a range of political messages beyond nationalism. These include showcasing the state's capabilities in responding to disasters and corruption, and promoting role models dedicated to advancing state objectives to motivate both officials and citizens. There are good reasons to believe that the effectiveness of propaganda varies across messages, as this experiment strongly indicates that audience preferences significantly impact its effectiveness. For instance, propaganda is more likely to backfire on opinion items related to extreme nationalism rather than benign patriotism, as well as on audiences who are predisposed to harbor antagonistic beliefs. But further research is needed to fully understand public receptivity to promoting different political themes beyond nationalism. If audience reception varies across different types of messaging, it would provide additional evidence that the state cannot shape popular opinion at will.

Future research is necessary to thoroughly understand why different audiences respond differently to the same messages. Plausible mechanisms include whether some perceive the propaganda effort to be unsettling or if they deem the situation worse considering the government's significant propaganda efforts. Moreover, further research is needed to fully explore how entertainment facilitates persuasion. For instance, it is worth investigating whether comedic elements are particularly conducive to persuasion.

References

- Adena, Maja et al. “Radio and the Rise of the Nazis in Prewar Germany”. In: *The Quarterly Journal of Economics* 130.4 (2015). Publisher: MIT Press, pp. 1885–1939 (cit. on pp. 2, 6, 9).
- Barber, Benjamin and Charles Miller. “Propaganda and combat motivation: Radio broadcasts and German soldiers’ performance in World War II”. In: *World Politics* 71.3 (2019), pp. 457–502 (cit. on p. 6).
- Barshad, Amos. “How Russian Action Movies Are Selling War”. en-US. In: *The New York Times* (Oct. 2022). <https://www.nytimes.com/2022/10/12/magazine/russian-action-movies.html>. ISSN: 0362-4331. (Visited on 10/24/2022) (cit. on p. 2).
- Baum, Matthew A. “Talking the vote: Why presidential candidates hit the talk show circuit”. In: *American Journal of Political Science* 49.2 (2005). Publisher: Wiley Online Library, 213–234 (cit. on p. 7).
- Baum, Matthew A. and Angela S. Jamison. “The Oprah effect: How soft news helps inattentive citizens vote consistently”. In: *The Journal of Politics* 68.4 (2006). Publisher: Cambridge University Press New York, USA, 946–959 (cit. on p. 7).
- Brady, Anne-Marie. *Marketing dictatorship: Propaganda and thought work in contemporary China*. Rowman & Littlefield Publishers, 2009 (cit. on p. 8).
- Chen, Yuyu and David Y. Yang. “The Impact of Media Censorship: 1984 or Brave New World?” In: *American Economic Review* 109.6 (June 1, 2019), pp. 2294–2332. DOI: [10.1257/aer.20171765](https://pubs.aeaweb.org/doi/10.1257/aer.20171765). URL: <https://pubs.aeaweb.org/doi/10.1257/aer.20171765> (cit. on p. 7).
- Coyne, Christopher J. and Abigail R. Hall. *Manufacturing Militarism: US Government Propaganda in the War on Terror*. Stanford University Press, 2021 (cit. on pp. 2, 35).
- El Banhawy, Yasmin. *Why is the Egyptian state monopolizing the entertainment industry?* en. <https://www.opendemocracy.net/en/north-africa-west-asia/why-egyptian-state-monopolizing-entertainment-industry/>. June 2019. (Visited on 07/04/2023) (cit. on p. 2).
- Ershad, Alijani. *Why Iran keeps making anti-American propaganda videos*. <https://observers.france24.com/en/20160831-iran-propaganda-music-videos-usa>. Aug. 2016. (Visited on 07/04/2023) (cit. on p. 2).

- Esarey, Ashley, Daniela Stockmann, and Jie Zhang. “Support for Propaganda: Chinese perceptions of public service advertising”. en. In: *Journal of Contemporary China* 26.103 (Jan. 2017), pp. 101–117. ISSN: 1067-0564, 1469-9400. DOI: [10.1080/10670564.2016.1206282](https://doi.org/10.1080/10670564.2016.1206282). URL: <https://www.tandfonline.com/doi/full/10.1080/10670564.2016.1206282> (visited on 02/16/2021) (cit. on p. 8).
- Esberg, Jane. “Censorship as Reward: Evidence from Pop Culture Censorship in Chile”. In: *American Political Science Review* 114.3 (Aug. 2020). Publisher: Cambridge University Press, pp. 821–836. DOI: [10.1017/S000305542000026X](https://doi.org/10.1017/S000305542000026X). URL: <https://www.cambridge.org/core/journals/american-political-science-review/article/censorship-as-reward-evidence-from-pop-culture-censorship-in-chile/1F2391F13BB0DACA6F844F6351FB5BA9> (cit. on p. 8).
- Frye, Timothy. *Weak Strongman: The Limits of Power in Putin’s Russia*. Princeton University Press, 2022 (cit. on p. 35).
- Gagliarducci, Stefano et al. “War of the waves: Radio and resistance during World War II”. In: *American Economic Journal: Applied Economics* 12.4 (2020), pp. 1–38 (cit. on p. 6).
- Gaines, Brian J. and James H. Kuklinski. “Experimental Estimation of Heterogeneous Treatment Effects Related to Self-Selection”. en. In: *American Journal of Political Science* 55.3 (2011). _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1540-5907.2011.00518.x>, pp. 724–736. ISSN: 1540-5907. DOI: [10.1111/j.1540-5907.2011.00518.x](https://doi.org/10.1111/j.1540-5907.2011.00518.x). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1540-5907.2011.00518.x> (visited on 10/14/2021) (cit. on p. 44).
- Green, Donald P. “In search of entertainment-education’s effects on attitudes and behaviors”. In: *Entertainment-Education Behind the Scenes*. Palgrave Macmillan, Cham, 2021, pp. 195–210 (cit. on p. 7).
- Green, Donald P., Dylan W. Groves, and Constantine Manda. “A Radio Drama’s Effects on Attitudes Toward Early and Forced Marriage: Results from a Field Experiment in Rural Tanzania”. In: *Unpublished Manuscript* (2020) (cit. on p. 7).
- Green, Donald P., Anna M. Wilke, and Jasper Cooper. “Countering violence against women by encouraging disclosure: A mass media experiment in rural Uganda”. In: *Comparative Political Studies* 53.14 (2020). Publisher: SAGE Publications Sage CA: Los Angeles, CA, pp. 2283–2320 (cit. on p. 7).

- Guriey, Sergei and Daniel Treisman. “A theory of informational autocracy”. In: *Journal of public economics* 186 (2020). Publisher: Elsevier, p. 104158 (cit. on p. 2).
- Han, Rongbin. “Manufacturing Consent in Cyberspace: China’s “Fifty-Cent Army””. en. In: *Journal of Current Chinese Affairs* 44.2 (June 2015). Publisher: SAGE Publications Ltd, pp. 105–134. ISSN: 1868-1026. DOI: [10.1177/186810261504400205](https://doi.org/10.1177/186810261504400205). URL: <https://doi.org/10.1177/186810261504400205> (visited on 04/15/2020) (cit. on p. 8).
- Hovland, Carl I, Arthur A Lumsdaine, and Fred D Sheffield. “Experiments on mass communication.(studies in social psychology in world war ii), vol. 3”. In: (1949) (cit. on p. 6).
- Hovland, Carl Iver, Irving Lester Janis, and Harold H. Kelley. “Communication and persuasion.” In: (1953). Publisher: Yale University Press (cit. on pp. 7, 8).
- Huang, Haifeng. “Propaganda as Signaling”. In: *Comparative Politics* 47.4 (July 2015), pp. 419–444. DOI: [10.5129/001041515816103220](https://doi.org/10.5129/001041515816103220) (cit. on p. 8).
- Huang, Haifeng. “The Pathology of Hard Propaganda”. In: *The Journal of Politics* 80.3 (May 2018). Publisher: The University of Chicago Press, pp. 1034–1038. ISSN: 0022-3816. DOI: [10.1086/696863](https://doi.org/10.1086/696863). URL: <https://www.journals.uchicago.edu/doi/full/10.1086/696863> (visited on 02/24/2021) (cit. on pp. 3, 6).
- Huang, Haifeng. “The Pathology of Hard Propaganda”. In: *The Journal of Politics* 80.3 (May 11, 2018). Publisher: The University of Chicago Press, pp. 1034–1038. DOI: [10.1086/696863](https://doi.org/10.1086/696863). URL: <https://www.journals.uchicago.edu/doi/full/10.1086/696863> (cit. on pp. 3, 8, 35).
- Kern, Holger Lutz and Jens Hainmueller. “Opium for the Masses: How Foreign Media Can Stabilize Authoritarian Regimes”. en. In: *Political Analysis* 17.04 (2009), pp. 377–399. ISSN: 1047-1987, 1476-4989. DOI: [10.1093/pan/mpp017](https://doi.org/10.1093/pan/mpp017). URL: https://www.cambridge.org/core/product/identifier/S1047198700012237/type/journal_article (visited on 10/04/2018) (cit. on pp. 6, 7).
- Kim, Eunji. *Entertaining Beliefs in Economic Mobility*. en. SSRN Scholarly Paper ID 3838127. Rochester, NY: Social Science Research Network, 2021. DOI: [10.2139/ssrn.3838127](https://doi.org/10.2139/ssrn.3838127). URL: <https://papers.ssrn.com/abstract=3838127> (visited on 09/26/2021) (cit. on pp. 7, 8).
- King, Gary, Jennifer Pan, and Margaret E. Roberts. “Reverse-engineering censorship in China: Randomized experimentation and participant observation”. In: *Science* 345.6199 (Aug. 2014). Publisher: American Association for the Advancement of Science, p. 1251722. DOI: [10.1126/](https://doi.org/10.1126/)

- science.1251722. URL: <https://www.science.org/doi/10.1126/science.1251722> (visited on 10/12/2021) (cit. on p. 8).
- Knobel, Beth. “The Great Game and the evolving nature of political talk shows on Russian television”. In: *Post-Soviet Affairs* 36.4 (July 2020). Publisher: Routledge _eprint: <https://doi.org/10.1080/1060586X.2020.1772664>, pp. 346–364. ISSN: 1060-586X. DOI: [10.1080/1060586X.2020.1772664](https://doi.org/10.1080/1060586X.2020.1772664). URL: <https://doi.org/10.1080/1060586X.2020.1772664> (visited on 09/14/2022) (cit. on p. 2).
- Koppes, Clayton R. and Gregory D. Black. “What to Show the World: The Office of War Information and Hollywood, 1942–1945”. In: *Journal of American History* 64.1 (1977), pp. 87–105 (cit. on p. 2).
- Little, Andrew T. “Propaganda and credulity”. In: *Games and Economic Behavior* 102 (2017), pp. 224–232 (cit. on p. 6).
- Lorentzen, Peter. “China’s Strategic Censorship”. In: *American Journal of Political Science* 58.2 (2014). Publisher: [Midwest Political Science Association, Wiley], pp. 402–414. URL: <https://www.jstor.org/stable/24363493> (cit. on p. 8).
- Mattingly, Daniel C. and Elaine Yao. “How soft propaganda persuades”. In: *Comparative Political Studies* (2022). Publisher: SAGE Publications Sage CA: Los Angeles, CA, p. 00104140211047403 (cit. on pp. 2, 3, 6–8, 16, 34, 35).
- Pan, Jennifer, Zijie Shao, and Yiqing Xu. “The Effects of Television News Propaganda: Experimental Evidence from China”. In: *Available at SSRN* (2020) (cit. on p. 8).
- Peisakhin, Leonid and Arturas Rozenas. “Electoral Effects of Biased Media: Russian Television in Ukraine: ELECTORAL EFFECTS OF BIASED MEDIA”. en. In: *American Journal of Political Science* 62.3 (July 2018), pp. 535–550. ISSN: 00925853. DOI: [10.1111/ajps.12355](https://onlinelibrary.wiley.com/doi/10.1111/ajps.12355). URL: <https://onlinelibrary.wiley.com/doi/10.1111/ajps.12355> (visited on 01/18/2022) (cit. on pp. 6, 9).
- Prior, Markus. “News vs. Entertainment: How Increasing Media Choice Widens Gaps in Political Knowledge and Turnout”. In: *American Journal of Political Science* 49.3 (2005). Publisher: [Midwest Political Science Association, Wiley], pp. 577–592. ISSN: 0092-5853. DOI: [10.2307/3647733](https://www.jstor.org/stable/3647733). URL: <https://www.jstor.org/stable/3647733> (visited on 02/17/2021) (cit. on p. 7).

- Prior, Markus. *Post-broadcast democracy: How media choice increases inequality in political involvement and polarizes elections*. Cambridge University Press, 2007 (cit. on p. 7).
- Repnikova, Maria. *Media politics in China: Improvising power under authoritarianism*. Cambridge University Press, 2017 (cit. on p. 8).
- Roberts, Margaret E. *Censored: distraction and diversion inside China's Great Firewall*. Princeton University Press, 2018 (cit. on p. 8).
- Sinaee, Maryam. *Iran's Regime Tightens Control Over Entertainment Media*. en. <https://www.iranintl.com/en/202306259077>. June 2023. (Visited on 07/04/2023) (cit. on p. 2).
- Sobolev, Anton. "How pro-government "trolls" influence online conversations in Russia". In: (2019) (cit. on p. 35).
- Soffar, Ossama. *The Egyptian TV Drama Taking Pro-Sisi Propaganda to New Heights*. en-US. <https://dawnmena.org/the-egyptian-tv-drama-taking-pro-sisi-propaganda-to-new-heights/>. Apr. 2022. (Visited on 07/04/2023) (cit. on p. 2).
- Stockmann, Daniela. *Media commercialization and authoritarian rule in China*. Cambridge University Press, 2013 (cit. on p. 8).
- Tolz, Vera and Yuri Teper. "Broadcasting agitainment: a new media strategy of Putin's third presidency". In: *Post-Soviet Affairs* 34.4 (July 2018). Publisher: Routledge _eprint: <https://doi.org/10.1080/1060586X.2018.1459023>, pp. 213–227. ISSN: 1060-586X. DOI: [10.1080/1060586X.2018.1459023](https://doi.org/10.1080/1060586X.2018.1459023). URL: <https://doi.org/10.1080/1060586X.2018.1459023> (visited on 09/14/2022) (cit. on p. 2).
- Voigtländer, Nico and Hans-Joachim Voth. "Nazi indoctrination and anti-Semitic beliefs in Germany". In: *Proceedings of the National Academy of Sciences* 112.26 (June 2015). Publisher: Proceedings of the National Academy of Sciences, pp. 7931–7936. DOI: [10.1073/pnas.1414822112](https://doi.org/10.1073/pnas.1414822112). URL: <https://www.pnas.org/doi/abs/10.1073/pnas.1414822112> (visited on 09/13/2022) (cit. on pp. 2, 6, 9).
- Wang, Tianyi. "Media, pulpit, and populist persuasion: Evidence from Father Coughlin". In: *American Economic Review* 111.9 (2021), pp. 3064–3092 (cit. on p. 6).
- Wang, Xiaohui. *Xin Shi Dai Wen Hua Jian She De Cheng Jiu Zhu Yao Bao Kuo Wu Ge Fang Mian [The Achievements of Cultural Construction in the New Era Mainly Include Five Aspects]*. 2021.

URL: <http://www.scio.gov.cn/xwfbh/xwbfbh/wqfbh/44687/47390/zy47394/Document/1716226/1716226.htm> (visited on 05/23/2023) (cit. on p. 36).

Wedeen, Lisa. “Acting “As If”: Symbolic Politics and Social Control in Syria”. In: *Comparative Studies in Society and History* 40.3 (1998), pp. 503–523. ISSN: 0010-4175. URL: <http://www.jstor.org/stable/179273> (visited on 04/02/2017) (cit. on p. 6).

Welch, David. *The Third Reich: politics and propaganda*. Routledge, 2008 (cit. on pp. 2, 6, 9).

A Appendix

A.1 Detailed Experimental Procedure

- (1) Baseline survey: If respondents chose to enter the study, they would first complete a short survey. In it, I asked them to select the movies they had seen from a list of films, and those who had seen the treatment or placebo movie were screened out. I then asked people about their demographic background, movie-watching preferences and habits, general satisfaction with the political and economic situation, and preferred choices between three pairs of movies. Regarding movie-watching preferences, participants were asked to choose from a list of genres, indicating their likes and dislikes. One of the available genre options was “main melody.” All surveys used in this experiment were hosted on Qualtrics, a reputable survey platform.
- (2) Random assignment and treatment delivery: I used *simple randomization* to assign each participant enrolled into one of three treatment conditions, with equal probabilities of enrollment in any of them. Then, each respondent received one online account providing a one-month free subscription to a movie website, IQiYi. IQiYi is the biggest streaming platform in China in terms of user count. It provides a diverse range of entertainment options, including the vast majority of major propaganda films. The account information came along with the encouragement to start with the movie recommended, according to their treatment assignment. They were also encouraged to continue to enjoy the movie subscription for as long as it remained valid (one month).

The treatment group was invited to view a propaganda movie, while the placebo control group was invited to view a nonpolitical placebo movie to serve as a baseline control group. In the free-choice group, people could choose to watch one of the two movies. The introduction of the free-choice group allowed us to take into account participants’ preferences for treatment options, a crucial consideration in understanding real-world effects shaped by having choices. This is also known as the patient preference trial (Gaines and Kuklinski, 2011).

- (3) Endline survey: One day after receiving the account information, respondents received the endline survey, and they were asked to fill out the survey only after they had watched the assigned or chosen movie. The survey included factual questions about the movie to determine

whether they had actually watched it carefully as well as inquiries about their general political attitudes and political engagement.

- (4) Tracking of viewing history: One month after enrolling a participant, I accessed their movie account and recorded their viewing history. This allowed me to confirm whether they had watched the recommended movie and what other movies they had watched.

A.2 Outcome Selection

Initially, I considered including more issue-specific opinions, such as hawkishness. However, these specific opinions were not included as primary outcomes in the final experiment. This decision was based on the lack of evidence from the pilot round indicating any significant impact of the movie on these attitudes. Additionally, it was observed that the movies did not significantly stimulate thinking related to any specific political issue, making it unlikely for them to affect more targeted political opinions.

In this study, nationalism encompasses both a benign attachment to one's country and the belief that one's country surpasses others in superiority. Economic perception represents participants' perceptions of the economic situation, including their own economic situation and the country's economic situation, both in the present and the future. System pride refers to whether individuals believe in their political system. Perceived government responsiveness reflects the public's perception of how well the incumbents are addressing their needs. Lastly, people's willingness to protest refers to their willingness to take to the streets, which is a significant concern for an authoritarian regime's survival.

A.3 Correlation matrices

These correlation matrices represent the correlations between the individual questions that combine to form an index for the outcome variables,

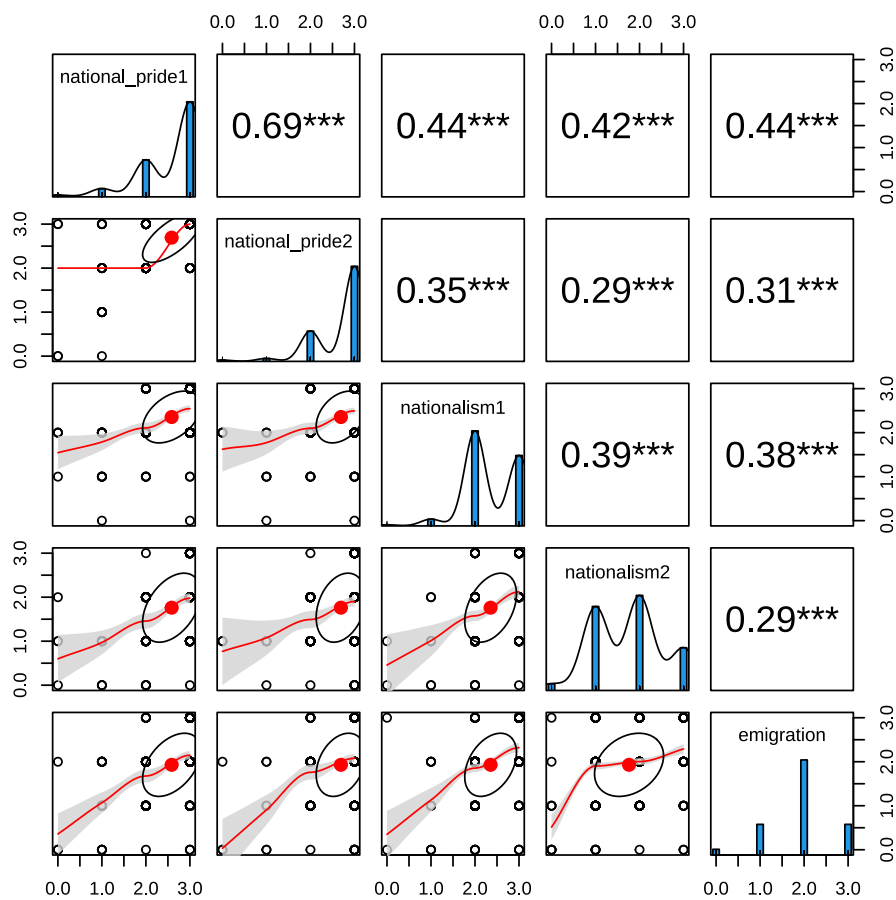


Figure 14: Correlational matrix: nationalism

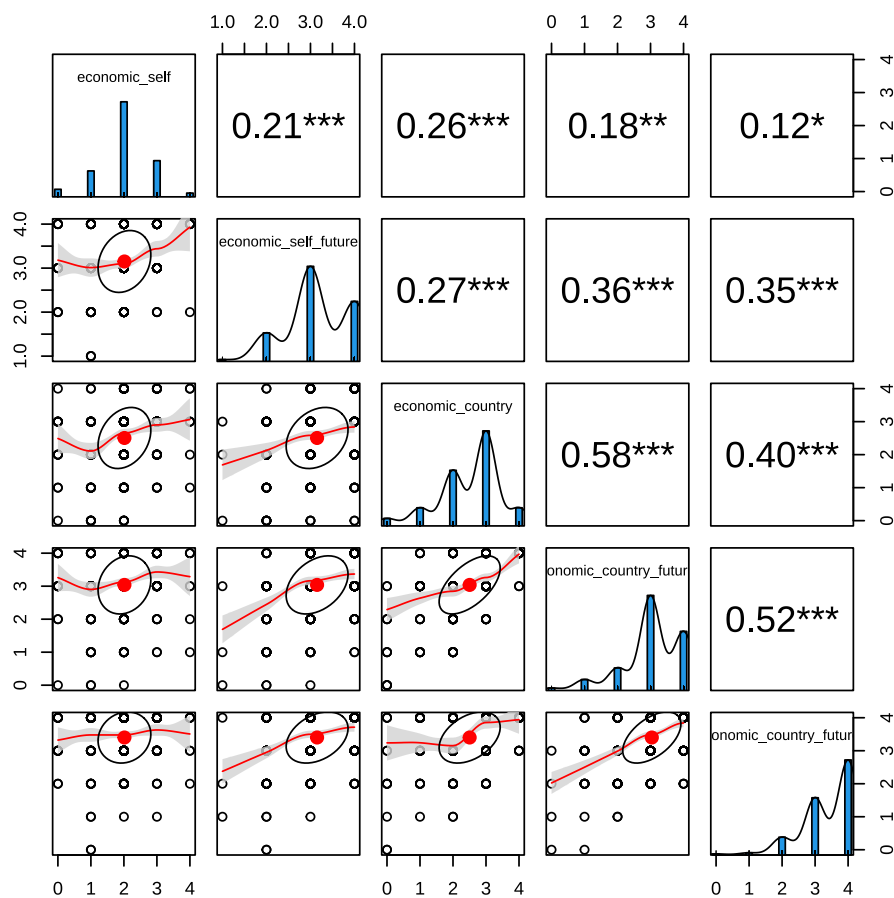


Figure 15: Correlational matrix: economic perception

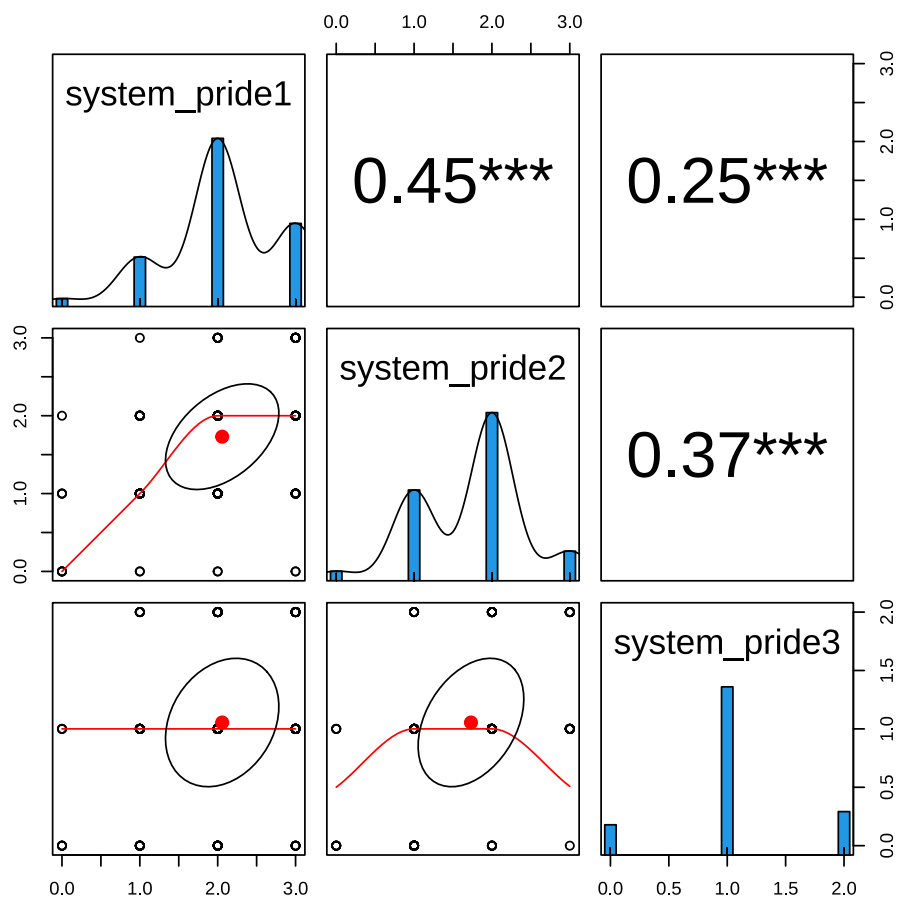


Figure 16: Correlational matrix: system pride

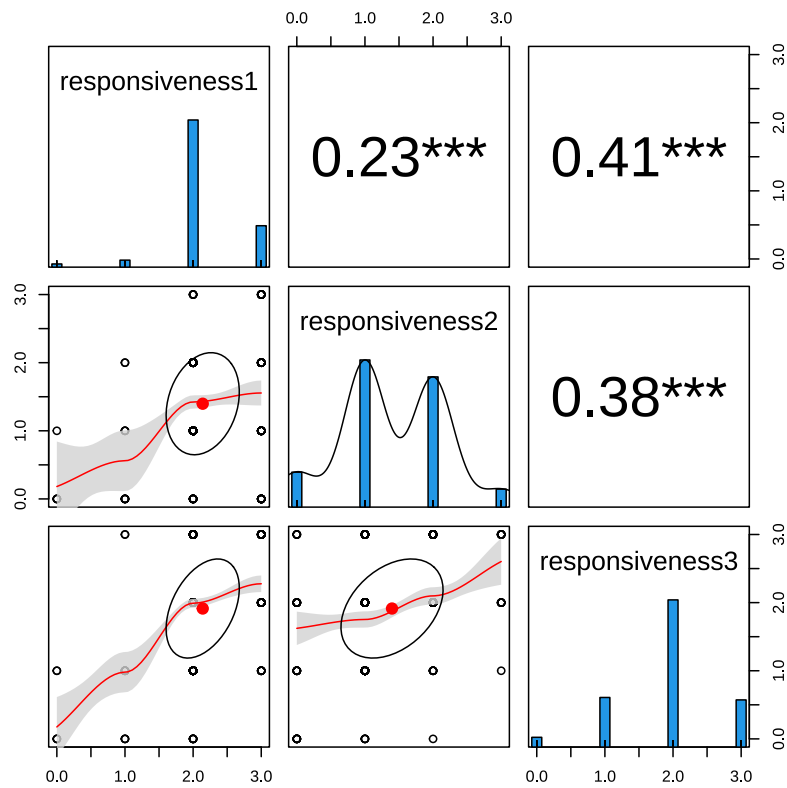


Figure 17: Correlational matrix: perceived responsiveness

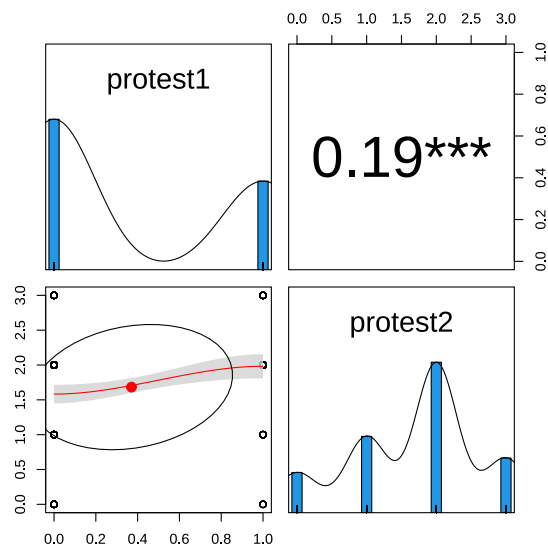


Figure 18: Correlational matrix: protest

A.4 Characteristics and Bias of Sample

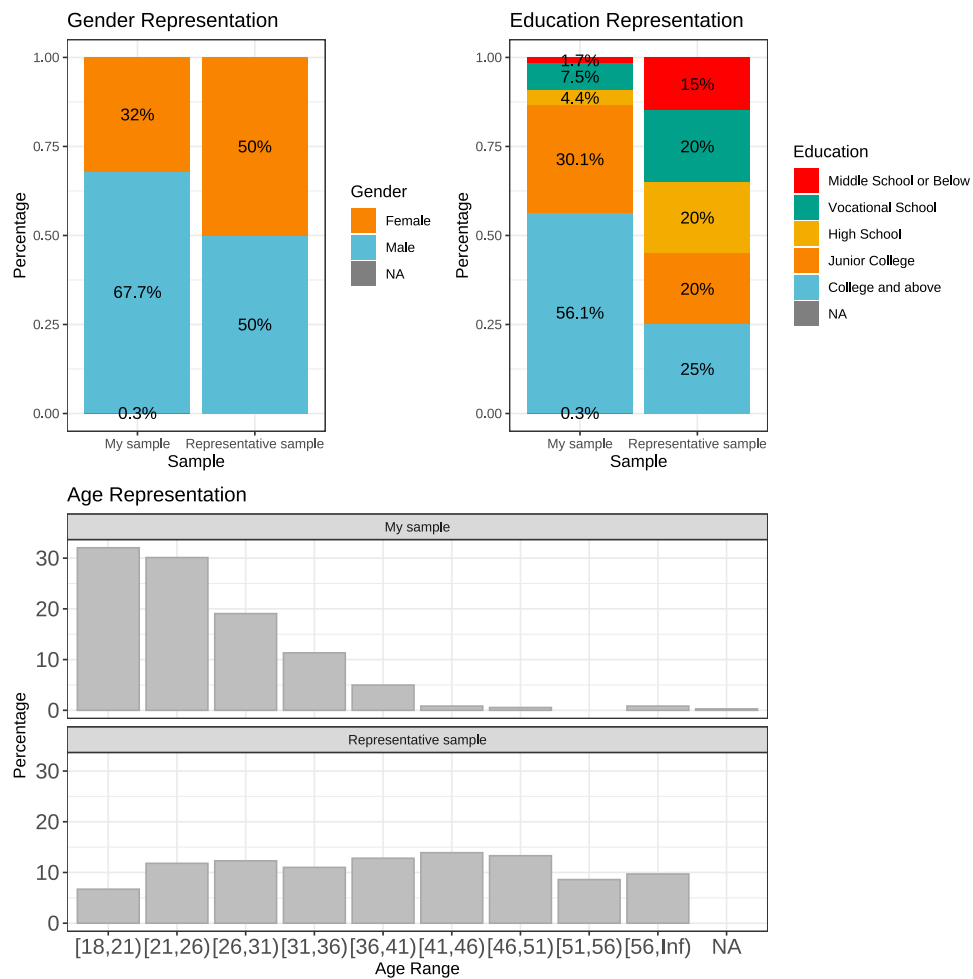


Figure 19: Characteristics and Bias of Sample

A.5 Sample Bias

Table 2: Sample Demographics

N	362.00
Age	
Mean	24.97
SD	6.65
Median	23.00
Min	18.00
Max	58.00
Gender (Count)	
Female	116
Male	245
Education (Count)	
High School or Lower	49
Three-year College	109
Bachelor's Degrees or Above	203
Residence (Count)	
First Tier City	89
Second Tier City	116
Third Tier City	81
Fourth Tier City	53
Rural Area	21

A.6 Who dislikes propaganda movies

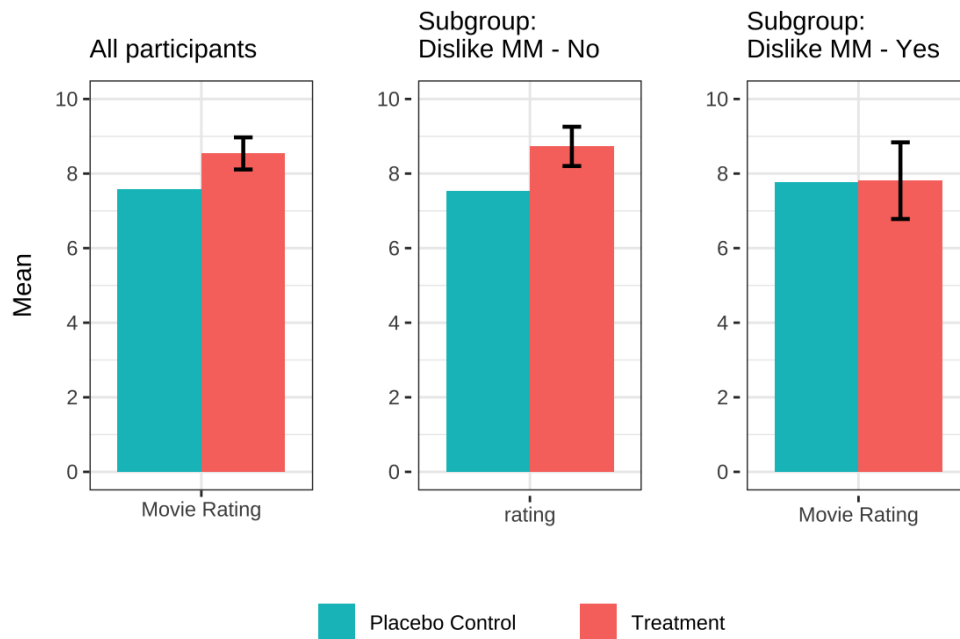
Table 3: Who Dislikes Propaganda Movies?

(Intercept)	0.474*
	(0.193)
Gender	−0.026
	(0.049)
Age	−0.002
	(0.004)
Education	−0.009
	(0.023)
City	−0.017
	(0.020)
Income	0.006
	(0.005)
Overall Satisfaction	−0.028
	(0.026)
Num.Obs.	360

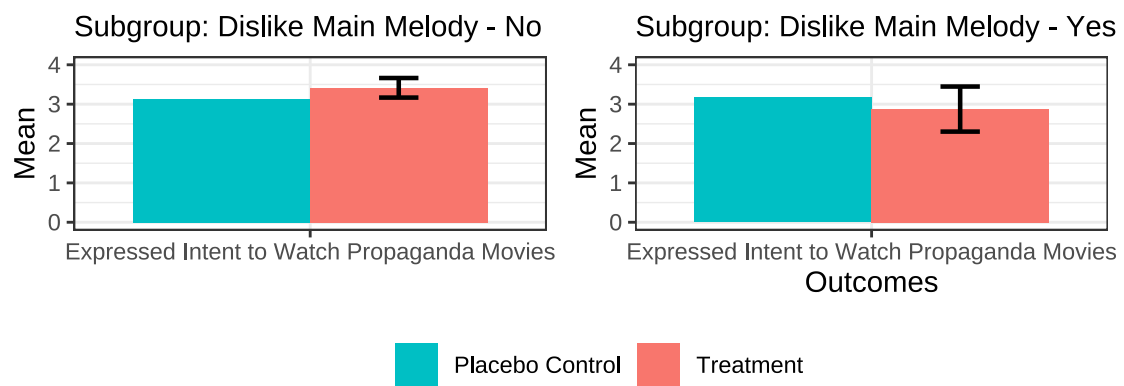
Table 4: Statistics: Movie Rating

Treatment Status	Mean	SD	N
All Participants			
Placebo	7.59	1.68	119
Treatment	8.54	1.55	100
Subgroup: Dislike MM - No			
Placebo	7.54	1.71	82
Treatment	8.78	1.37	76
Subgroup: Dislike MM - Yes			
Placebo	7.70	1.63	37
Treatment	7.79	1.86	24

A.7 Movie Ratings



A.8 Expressed intent to watch



	Index	Questions	Scales
1	intent_propaganda	Intent to Watch Wolf Warrior 3	0 = Definitely No, 3 = Definitely Yes

Figure 20: Expressed intent to watch propaganda

A.9 Audience's Taste in Entertainment by Treatment Group

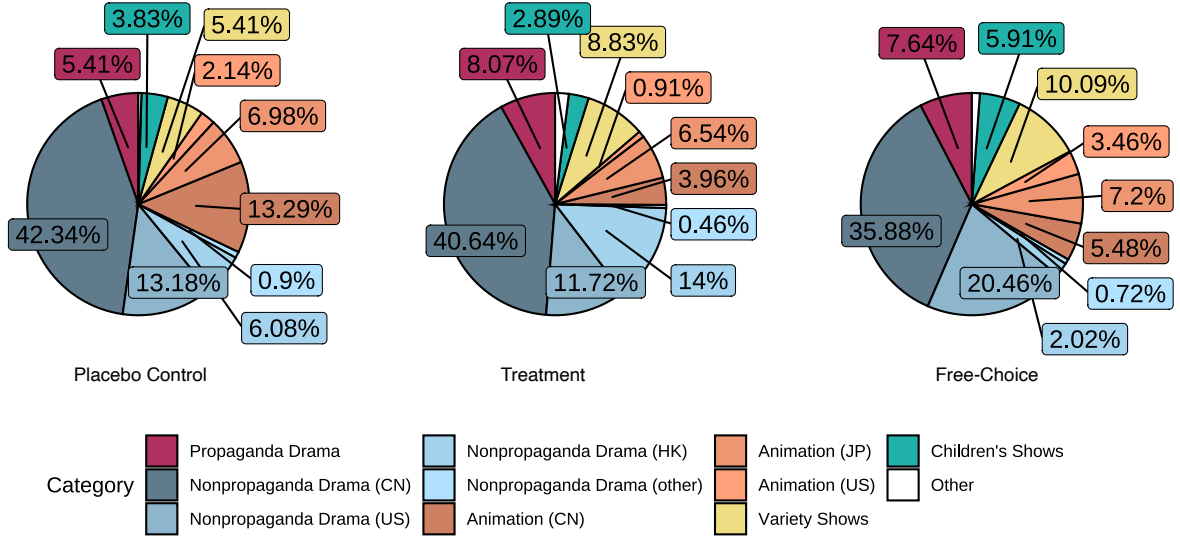


Figure 21: Audience's Taste in Entertainment by Treatment Group

A.10 Who Chooses the Propaganda Movie in the Free-choice Group

Table 5: Who Chooses the Propaganda Movie?

(Intercept)	0.741 (0.544)
Gender	-0.097 (0.108)
Age	-0.010 (0.010)
Education	0.082 (0.058)
City	-0.050 (0.048)
Income	-0.014 (0.013)
Overall Satisfaction	0.033 (0.065)
Num.Obs.	92

Table 6: Including the Free-Choice Group: Number of Subjects and Probability of Watching Propaganda

Treatment Group	Number of Subjects	Probability of Watching the Propaganda Movie
placebo	139	0.00
free-choice	107	0.63
treatment	116	1.00

Table 7: Including the Free-Choice Group: Emotional Response to Movies

Treatment Group	Happiness		Sadness		Pride		Anger	
	mean	sd	mean	sd	mean	sd	mean	sd
placebo	2.52	0.86	2.80	0.91	2.52	1.20	2.52	1.03
free-choice	2.84	0.87	2.74	0.88	2.84	1.48	2.84	1.02
treatment	2.83	0.95	2.87	0.88	2.83	1.02	2.83	0.90

Table 8: Including the Free-Choice Group: Political Opinions

Treatment Group	Nationalism		Economic Perception		System Pride		Responsiveness		Protest	
	mean	sd	mean	sd	mean	sd	mean	sd	mean	sd
placebo	-0.05	0.89	-0.11	0.64	-0.05	0.76	-0.05	0.69	-0.06	0.75
free-choice	-0.02	0.91	0.05	0.60	-0.03	0.71	-0.05	0.81	0.10	0.79
treatment	0.06	0.92	0.07	0.77	0.08	0.79	0.11	0.76	-0.01	0.78

Table 9: Including the Free-Choice Group: Consumption of the Propaganda Movie in the Subsequent Month

Treatment Group	Number of Propaganda Movies Watched	
	mean	sd
placebo	0.49	1.68
free-choice	0.53	0.93
treatment	0.55	0.74

A.11 ITT effects

Table 10: ITT effects

term	Nationalism	Economic Perception	System Pride	Responsiveness	Protest
(Intercept)	−0.052 (0.083)	−0.107+ (0.059)	−0.052 (0.070)	−0.053 (0.063)	−0.062 (0.069)
Treatment: Forced Exposure	0.114 (0.123)	0.175+ (0.097)	0.137 (0.105)	0.166+ (0.098)	0.055 (0.104)
Treatment: Free-Choice	0.036 (0.125)	0.158+ (0.086)	0.026 (0.102)	0.006 (0.105)	0.160 (0.107)
Num.Obs.	309	311	312	312	312

A.12 Self-selectors and non-selectors

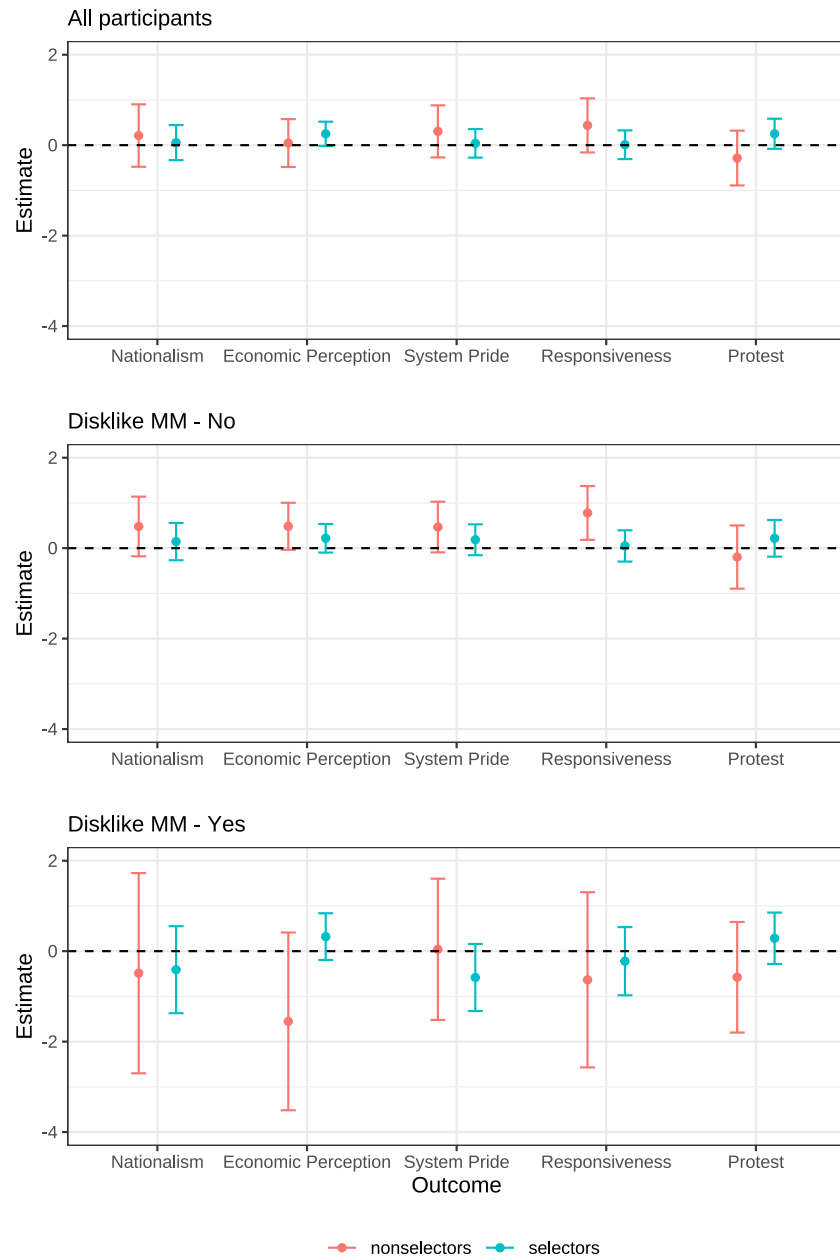


Figure 22: Estimated Treatment Effects Among Selectors and Nonselectors

A.13 Compliance and CACEs

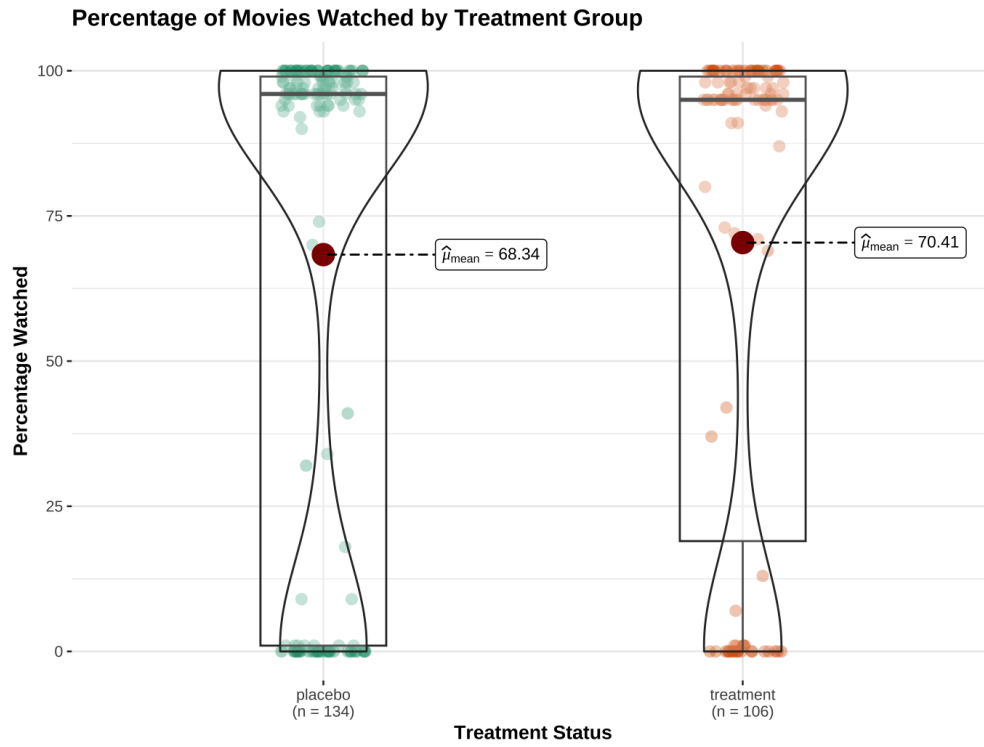


Figure 23: Percentage of Movies Watched by Treatment Group

Table 11: Statistics: Percentage of Movies Watched

treatment_status	group	n	percentage
placebo	watched more than 90 percent	90	67.16
treatment	watched more than 90 percent	71	66.98
placebo	did not watch the movie	16	11.94
treatment	did not watch the movie	12	11.32
placebo	account not used	19	14.18
treatment	account not used	13	12.26

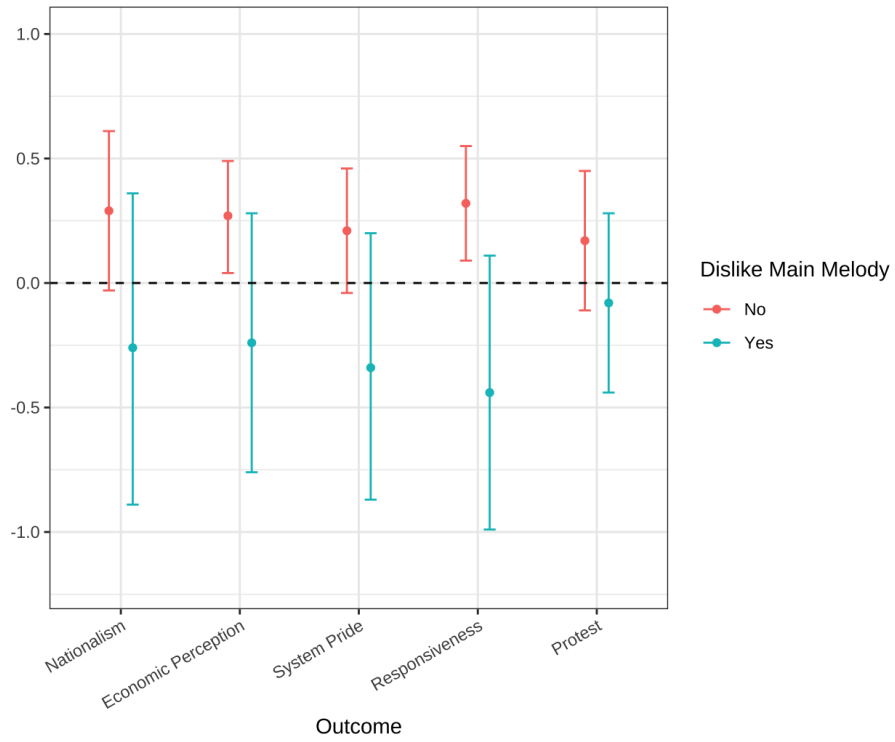
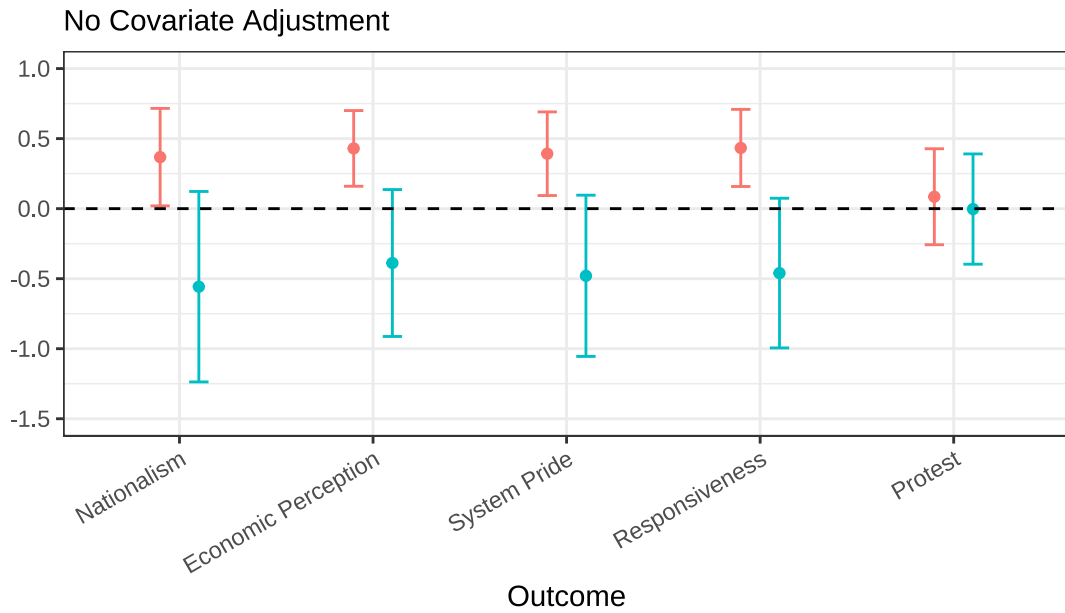
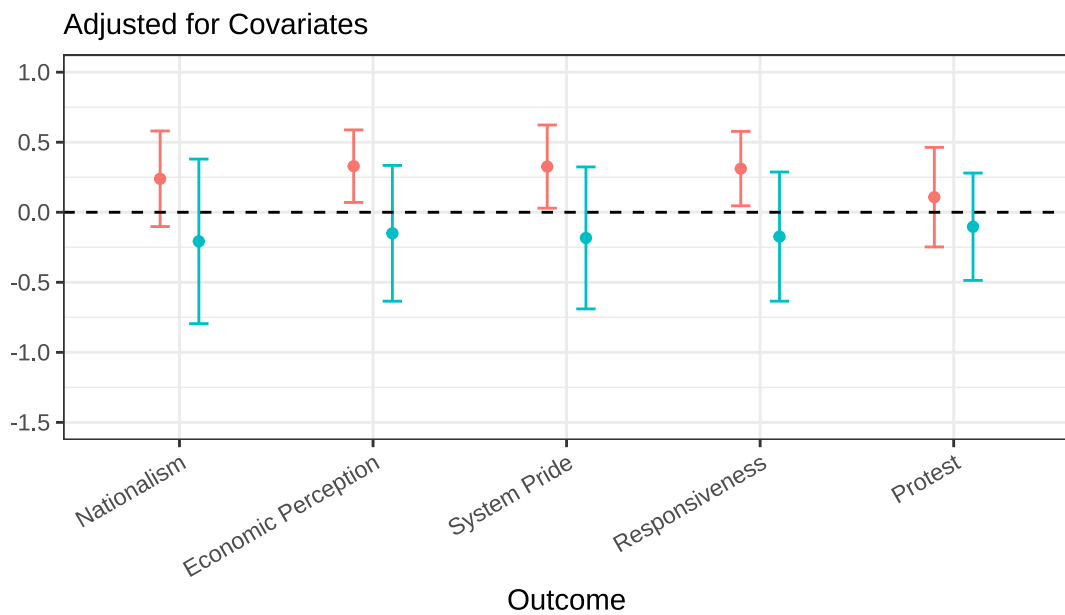


Figure 24: Estimated CACEs: Using Percentage Watched as Measures of Compliance



Dislike Main Melody — No — Yes



Dislike Main Melody — No — Yes

Note: Covariates include gender, age, education, city, income, overall satisfaction.

Figure 25: Estimated CACEs: Using Attention Check Question as Measures of Compliance

A.14 Covariate Imbalance

Table 12: Balance of covariates

	Gender	Age	Education	City	Income	Party Af- filiation	Overall Satisfac- tion
<i>Panel 1: All Participants</i>							
(Intercept)	1.302*** (0.040)	24.633*** (0.584)	4.309*** (0.092)	2.288*** (0.100)	4.813*** (0.401)	2.079*** (0.091)	3.266*** (0.078)
Treatment	0.046 (0.059)	0.263 (0.869)	0.099 (0.136)	0.277+ (0.148)	-1.109+ (0.595)	-0.123 (0.135)	0.134 (0.116)
Num.Obs.	254	254	254	254	254	254	254
<i>Panel 2: Dislike MM - No</i>							
(Intercept)	1.303*** (0.048)	24.838*** (0.711)	4.253*** (0.107)	2.323*** (0.121)	4.404*** (0.446)	2.030*** (0.108)	3.212*** (0.092)
Treatment	0.075 (0.069)	-0.272 (1.030)	0.247 (0.155)	0.266 (0.175)	-0.760 (0.646)	-0.053 (0.156)	0.277* (0.133)
Num.Obs.	189	189	189	189	189	189	189
<i>Panel 3: Dislike MM - Yes</i>							
(Intercept)	1.300*** (0.072)	24.125*** (1.008)	4.450*** (0.178)	2.200*** (0.174)	5.825*** (0.865)	2.200*** (0.173)	3.400*** (0.143)
Treatment	-0.060 (0.116)	1.955 (1.625)	-0.370 (0.287)	0.280 (0.281)	-1.905 (1.395)	-0.320 (0.278)	-0.320 (0.230)
Num.Obs.	65	65	65	65	65	65	65

A.15 Covariate Adjustment

Table 13: Effects of Propaganda Movie on Opinions: Adjusted for Covariates, All Participants

	Nationalism	Economic Perception	System Pride	Perceived Responsive- ness	Willingness to Protest
(Intercept)	−0.795 (0.500)	−0.788* (0.358)	−0.930* (0.408)	−1.031** (0.366)	0.588 (0.430)
Treatment	0.071 (0.127)	0.137 (0.089)	0.109 (0.102)	0.131 (0.091)	0.082 (0.107)
Gender	0.025 (0.131)	0.182+ (0.093)	0.103 (0.106)	0.153 (0.095)	−0.107 (0.111)
Age	−0.012 (0.010)	−0.016* (0.007)	−0.006 (0.008)	−0.012 (0.008)	−0.013 (0.009)
Education	−0.051 (0.059)	−0.061 (0.043)	−0.021 (0.049)	0.000 (0.044)	0.015 (0.052)
City	−0.015 (0.056)	−0.017 (0.039)	−0.031 (0.044)	−0.033 (0.040)	−0.030 (0.046)
Income	0.000 (0.015)	−0.002 (0.011)	0.010 (0.012)	0.002 (0.011)	−0.003 (0.013)
Party Affiliation		0.064 (0.042)	0.001 (0.048)	0.059 (0.043)	0.013 (0.051)
Overall Satisfaction	0.378*** (0.068)	0.303*** (0.048)	0.302*** (0.055)	0.308*** (0.049)	−0.060 (0.058)
Num.Obs.	217	219	219	219	219

Table 14: Effects of Propaganda Movie on Opinions: Adjusted for Covariates, Dislike MM - No

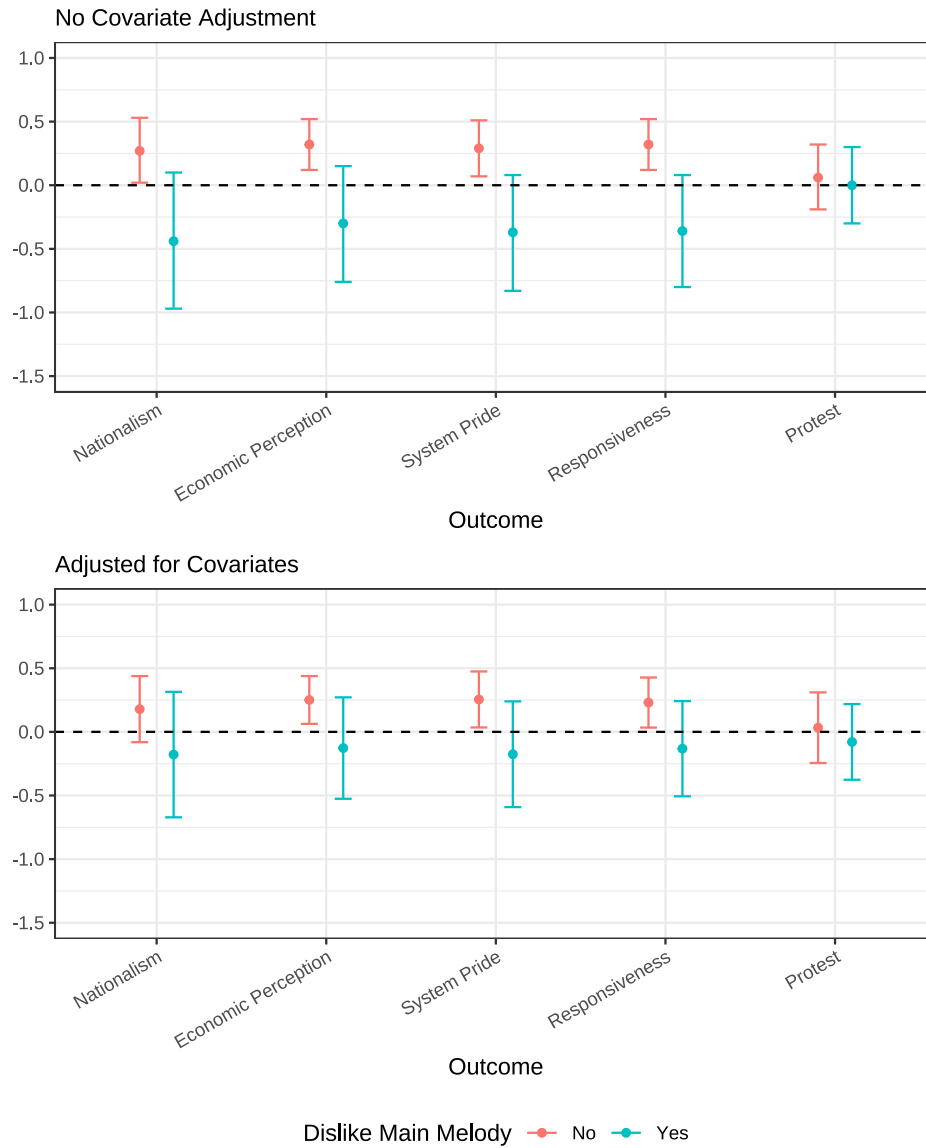
	Nationalism	Economic Perception	System Pride	Perceived Responsive- ness	Willingness to Protest
(Intercept)	−0.488 (0.540)	−0.387 (0.375)	−0.520 (0.439)	−0.798* (0.388)	0.482 (0.531)
Treatment	0.234 (0.142)	0.257** (0.098)	0.247* (0.115)	0.241* (0.101)	0.085 (0.139)
Gender	−0.030 (0.142)	0.135 (0.098)	0.071 (0.115)	0.100 (0.101)	−0.034 (0.139)
Age	0.003 (0.012)	−0.013 (0.008)	0.004 (0.010)	−0.005 (0.008)	−0.018 (0.012)
Education	−0.091 (0.067)	−0.114* (0.048)	−0.077 (0.056)	−0.045 (0.050)	−0.002 (0.068)
City	0.007 (0.060)	−0.023 (0.040)	−0.030 (0.047)	−0.011 (0.042)	−0.030 (0.057)
Income	−0.013 (0.020)	0.003 (0.013)	0.010 (0.015)	0.007 (0.013)	0.001 (0.018)
Party Affiliation		0.089+ (0.046)	−0.013 (0.054)	0.054 (0.048)	0.041 (0.065)
Overall Satisfaction	0.259*** (0.076)	0.241*** (0.053)	0.210*** (0.062)	0.257*** (0.054)	−0.015 (0.074)
Num.Obs.	157	159	159	159	159

Table 15: Effects of Propaganda Movie on Opinions: Adjusted for Covariates, Dislike MM - Yes

	Nationalism	Economic Perception	System Pride	Perceived Responsive- ness	Willingness to Protest
(Intercept)	−0.528 (1.173)	−1.151 (0.977)	−1.405 (1.001)	−0.477 (0.918)	1.687* (0.764)
Treatment	−0.287 (0.261)	−0.119 (0.201)	−0.143 (0.206)	−0.135 (0.189)	−0.088 (0.157)
Gender	−0.129 (0.296)	0.117 (0.228)	−0.061 (0.233)	0.069 (0.214)	−0.407* (0.178)
Age	−0.027 (0.023)	−0.016 (0.018)	−0.015 (0.019)	−0.029+ (0.017)	−0.012 (0.014)
Education	−0.016 (0.120)	0.010 (0.092)	0.086 (0.095)	0.048 (0.087)	0.029 (0.072)
City	−0.166 (0.125)	−0.052 (0.099)	−0.076 (0.101)	−0.159+ (0.093)	−0.106 (0.077)
Income	0.003 (0.026)	−0.013 (0.020)	−0.003 (0.021)	−0.008 (0.019)	0.003 (0.016)
Party Affiliation		0.014 (0.095)	0.042 (0.097)	0.049 (0.089)	−0.077 (0.074)
Overall Satisfaction	0.483** (0.157)	0.392** (0.121)	0.426** (0.124)	0.303* (0.113)	−0.213* (0.094)
Num.Obs.	60	60	60	60	60

A.16 Covariate Adjustment: Graph

The figure below presents a comparison of the estimates with and without covariate adjustment. As the treatment groups were unbalanced, the estimated effects were found to be smaller after controlling for covariates. However, the overall pattern of results remained consistent.



Note: Covariates include gender, age, education, city, income, party affiliation, and overall satisfaction.

Figure 26: No Covariate Adjustment Versus Adjusted for Covariate

A.17 Estimated Heterogeneous Treatment Effects By Overall Satisfaction

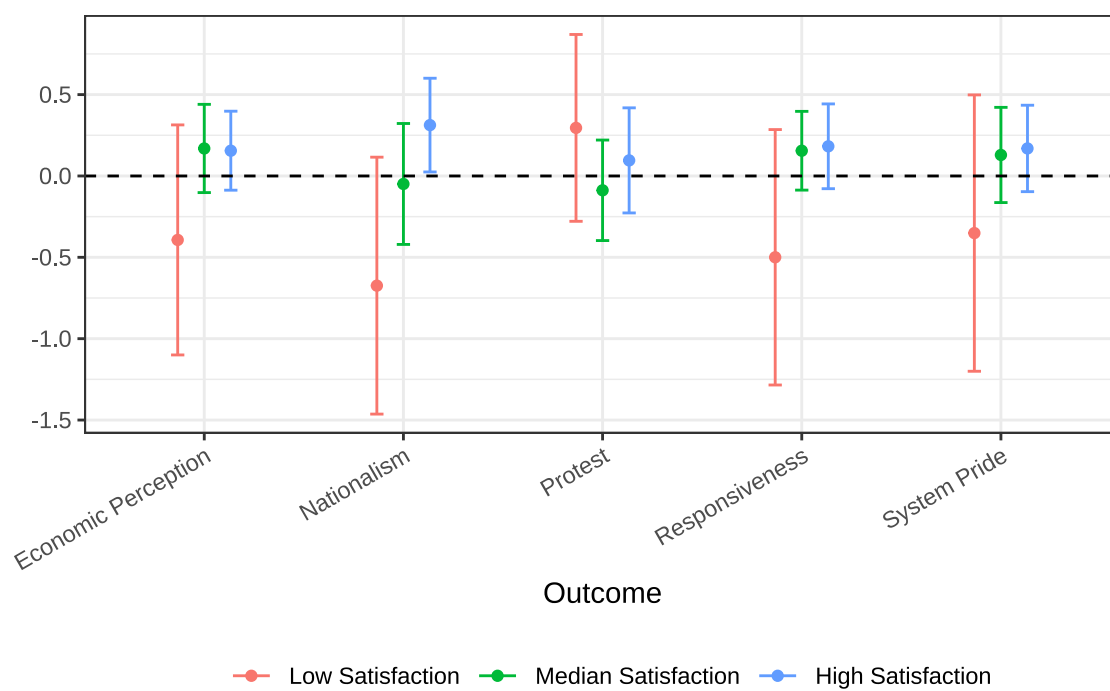


Figure 27: Estimated Heterogeneous Treatment Effects By Overall Satisfaction

A.18 Estimated Heterogeneous Treatment Effects By Movie-Watching Frequency

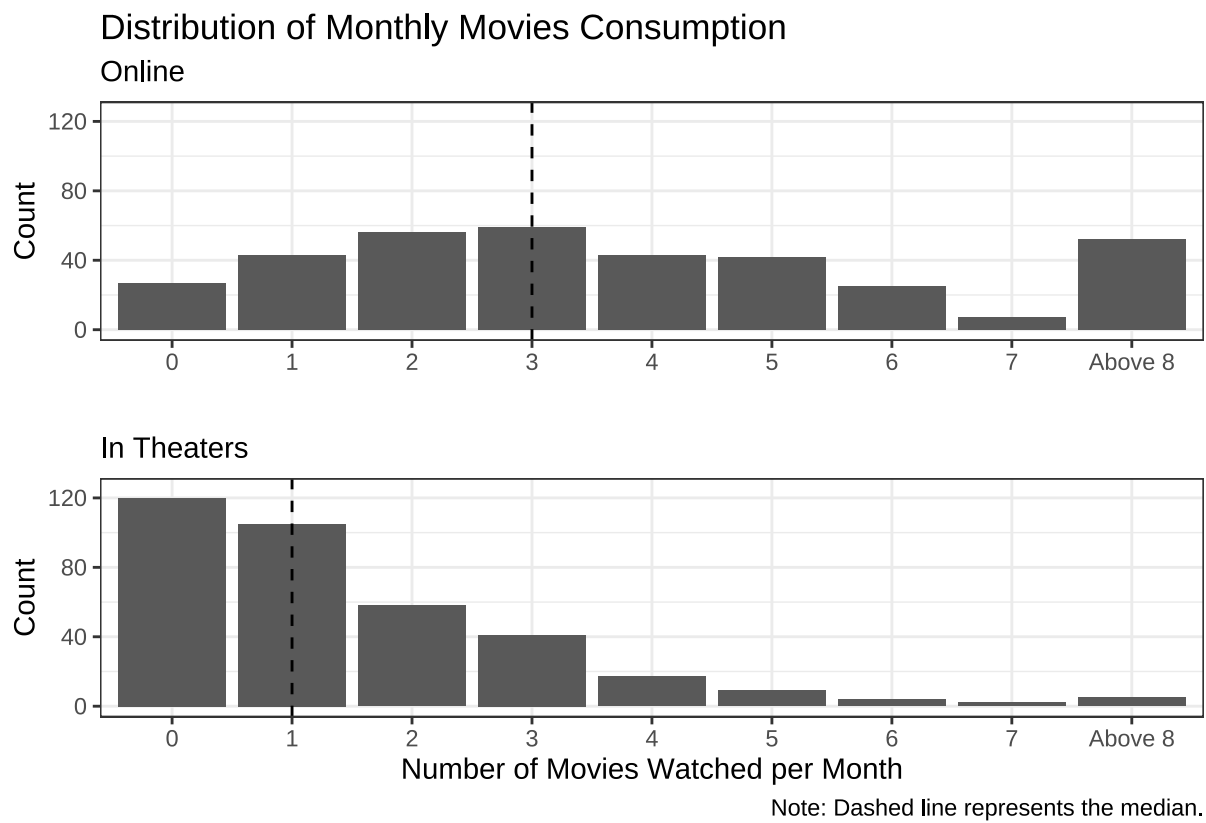
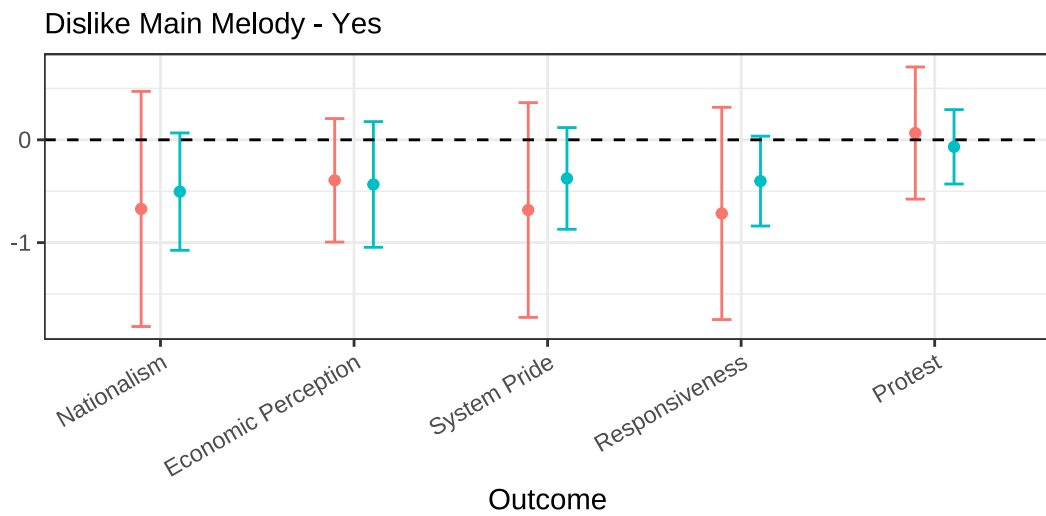
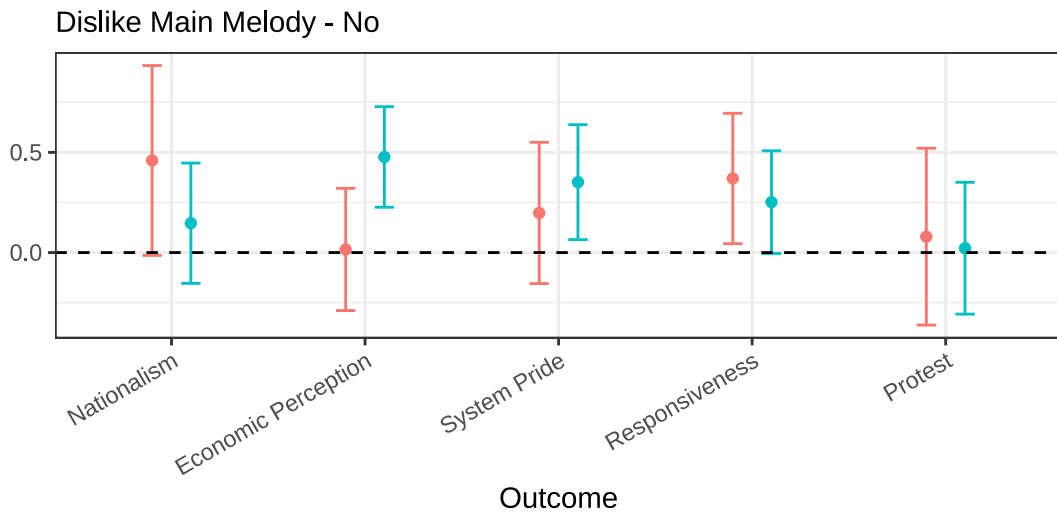
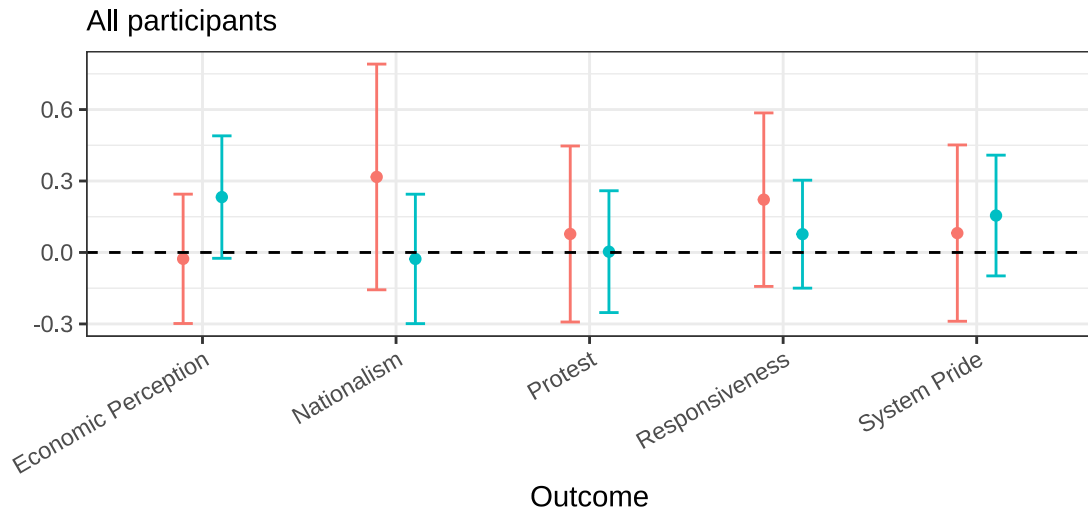


Figure 28: Distribution of Movie-Watching Frequency



Frequent Online Movie Watchers — No — Yes

Figure 29: Estimated Treatment Effects By Online Movie-Watching Frequency

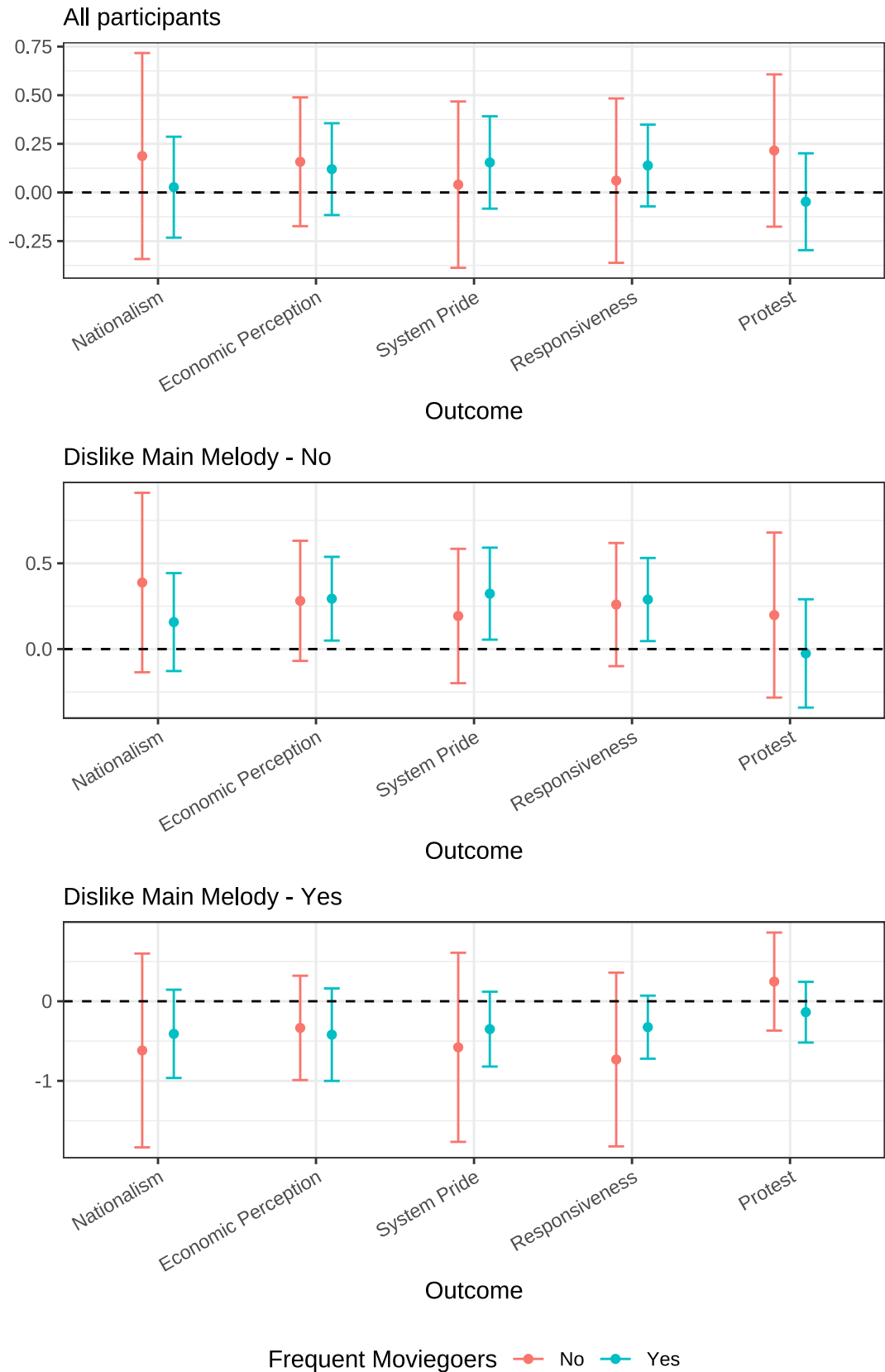


Figure 30: Estimated Treatment Effects By Movie-Going Frequency

A.19 Estimated Heterogeneous Treatment Effects By Gender

One issue that complicates this exercise is the small number of female participants who expressed a distaste for propaganda. This means that the estimates for this subgroup are particularly imprecise. Other than that, some minor gender differences are observed, but the overall pattern remains roughly similar.

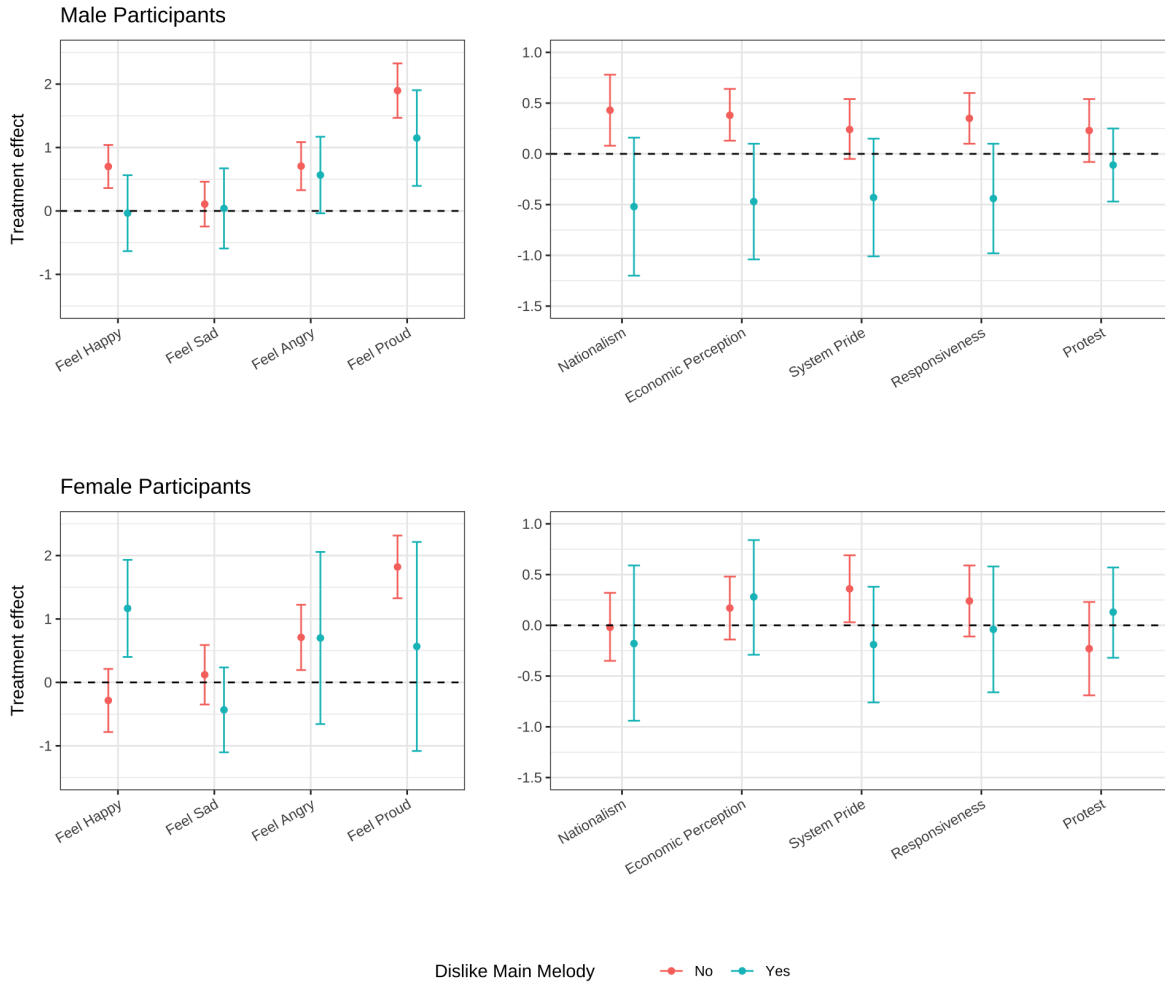


Figure 31: Estimated Treatment Effects By Gender

Table 16: Gender: Sample Bias

Gender	Dislike MM	N
Male	Yes	183
Male	Yes	62
Female	Yes	91
Female	Yes	25